Two Hidden Layers Suffice: The Caterpillar Net

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Abstract

Every continuous piecewise linear (CPWL) function is a finite signed sum of maxima of affine functions, making maximum representation a central problem in the study of the expressiveness of ReLU networks. Earlier work disproved the conjectured base-two depth bound, leaving open whether a constant number of hidden layers suffices to represent the maximum. Independently and concurrently, Brunck, Hertrich, Rueß, Stade, and Yehudayoff established that two hidden layers suffice for all CPWL functions. We give the *Caterpillar Net*, a non-recursive explicit construction representing the maximum of N real inputs exactly with two hidden layers of widths $\binom{N}{2} + N + 1$ and $2^{O(N \log N)}$, rational weights, and zero biases. Every expansion of the maximum in its finite vocabulary compiles with the same first hidden layer. Sharing its first hidden layer across a Wang–Sun representation gives two-hidden-layer networks for all CPWL functions. [1]

1 Introduction

In this paper we investigate the representation capabilities of ReLU networks: what functions are representable by such networks of different depths and widths? ReLU networks represent precisely the continuous piecewise-linear (CPWL) functions with finitely many pieces [ABMM18]. A sharp question is therefore: given a CPWL function, what is the smallest depth, or the smallest width at a prescribed depth, needed to represent it exactly [HBD+23, Saf26]? The maximum is a universal building block for such functions: every CPWL function is a finite signed sum of maxima of affine functions [WS05]. Write

$$\max_N(x) := \max_{0 \le i < N} x_i. \tag{1}$$

Theorem (Wang–Sun, [WS05], `Lean_1`)**.** *Let $d \ge 1$ and let $f : \mathbb{R}^d \to \mathbb{R}$ be continuous and affine on each member of a finite polyhedral cover. There are $R \in \mathbb{N}$, affine maps and signs*

$$A_t : \mathbb{R}^d \longrightarrow \mathbb{R}^{d+1}, \qquad \varepsilon_t \in \{-1, 1\} \qquad (1 \le t \le R) \tag{2}$$

such that

$$f(x) = \sum_{t=1}^{R} \varepsilon_t \max_{d+1}(A_t(x)) \qquad \textit{for every } x \in \mathbb{R}^d. \tag{3}$$

[1] Our results were obtained over months of proof search in Lean using Claude Code and Codex; the AI methodology statement describes their roles. The Lean code is available in the accompanying repository https://github.com/JayPiZimmermann/Max_Caterpillar. Links to the Lean formalization appear in theorem titles and equation numbers; equation numbers are clickable.

Thus a two-hidden-layer representation of $\max_{d+1}$ gives one for every CPWL function on $\mathbb{R}^d$: precompose with the affine maps and add the networks in parallel. Section 8 reuses their first hidden layer; the representation theorems and the resulting networks are formalized.

Our contribution. The *Caterpillar Net* gives a nonrecursive two-hidden-layer representation of the maximum with explicit bounds on both hidden-layer widths. The construction uses a finite dictionary of support functions of caterpillar polytopes and their faces. The maximum is the support function of a simplex. We first express this support function as an alternating sum of caterpillar support functions, up to a linear combination of support functions of lower-dimensional polytopes. We then represent the remaining error using faces of the same caterpillar polytopes and elementary ReLU features. Normalizing and merging these descriptions yields a width bound for the complete network, including all correction terms.

Write $\sigma(t) := \max(0, t)$ applied coordinatewise to vectors. A network has an input layer of N coordinates, first and second hidden layers of widths h_1 and h_2, respectively, and a scalar output layer. With real weight matrices W_1, W_2, output vector w_3, and biases b_1, b_2, b_3 of the corresponding dimensions, it evaluates as

$$x \longmapsto w_3^\top \sigma\big(W_2 \sigma(W_1 x + b_1) + b_2\big) + b_3. \tag{4}$$

There are no skip connections; equality is required on all of $\mathbb{R}^N$. For $N = 1, 2, 3$ there are known simple network representations of width $(2, 2)$ and $(4, 2)$ and $(5, 3)$; the last nests $\max(s, t) = t + \sigma(s - t)$. Our Main Theorem presents such constructions for higher N. A network has *rational weights* if every entry of W_1, W_2, and w_3 is rational, and *zero biases* if b_1, b_2, and b_3 vanish.

Theorem 1 (Caterpillar Net, `Lean_2`, Proof)**.** *For every $N \geq 3$, there is a two-hidden-layer ReLU network with rational weights and zero biases computing $\max_N$ with widths*

$$h_1 = \binom{N}{2} + N + 1, \qquad h_2 = 3^{N-2} C_N - N(N-1) + 2, \tag{5}$$

where

$$C_N := \sum_{m=2}^{N-1} \binom{N}{m} \frac{m!}{2} (N - m) m^{N-m-1}. \tag{6}$$

Thus $h_1 = O(N^2)$ and $h_2 = 2^{O(N \log N)}$ as $N \to \infty$.

Further results. The network is compiled from an expansion of $\max_N$ in a finite vocabulary of caterpillar face functions. Theorem 6 separates the network from the coefficients: any such expansion with $|K|$ retained units compiles with the same first hidden layer and $|K| + 2$ second hidden layer neurons. The coefficients are the coordinates of $\max_N$ in one basis of the vocabulary; the same proof shows that they are rational, without any integral (Theorem 14). Section 7 bounds the second width by $q_N + 2$, where q_N is the dimension of the vocabulary modulo its auxiliary functions; this bound proves Theorem 1. For $N = 3, 4$ expansions with one and three units are proved in Lean. Conversely, Section 4 proves that every expansion of $\max_N$ in caterpillar and auxiliary functions has at least $(7/6)^N/(N + 1)$ caterpillar functions, and that $q_N \geq (N - 2)!$. Section 8 applies the construction to sums of maxima.

2 Related work

The conjecture that $\max\{0, x_1, \ldots, x_{2^k}\}$ requires more than k hidden layers [HBD+23] was disproved for $k = 2$ by the maximum of five inputs in [BBH+26]. Exact computer-assisted searches subsequently obtained

two-hidden-layer representations up to $N = 12$ in [RAB+26] and, independently, up to $N = 8$ in [WB26]. The improved general depth bounds remained logarithmic, leaving constant depth open.

Setting all biases to zero does not change a network that computes a positively homogeneous function such as $\max_N$ [HBD+23, Proposition 2.3], [HHL23, Lemma 7], so biases give no additional power for representing the maximum. Depth lower bounds hold for integer weights [HHL23] and rational weights whose denominators are restricted to powers of a fixed integer [AHM25]. Our weights are rational, but their denominators are not restricted in this way, so neither bound applies. Other results concern braid-fan-compatible networks [GHL25] and first hidden layer width bounds [Saf26, Lu26].

Theorem 2 (Brunck, Hertrich, Rueß, Stade, and Yehudayoff, [BHR+26])**.** *Every continuous piecewise affine function $f : \mathbb{R}^d \to \mathbb{R}$ with finitely many affine pieces can be represented exactly by a ReLU network with two hidden layers.*

Their first proof proceeds by induction on dimension, using homological algebra to construct a linear combination of support functions of pyramids over zonotopes that agrees with the simplex support function modulo lower-dimensional terms. Our construction restricts these building blocks to an explicitly specified finite family, the caterpillar polytopes; we deliver a geometric proof of the corresponding identity.

Independently and concurrently, they developed a second proof based on combinatorial cancellation, with building blocks closely related to ours. Appendix D identifies them exactly: for nonempty marked sets their pyramids are translated caterpillar polytopes, and the remaining pyramids are rooted stars, which are not caterpillar functions. We do not claim that their coefficients agree with ours. We establish explicit width bounds for the first and second hidden layers; their paper states no widths.

3 The Caterpillar Net

Two identities drive the construction:

$$\max(s, t) = t + \sigma(s - t), \tag{7}$$

$$\sigma(\ell(x)) = \sigma(-\ell(x)) + \ell(x). \tag{8}$$

The first suggests a path with attached leaves. A signed sum of the resulting caterpillar support functions agrees with the maximum modulo lower-dimensional terms; proper faces and linear functions correct the error. The second identity reverses hinges to merge units in Section 6.

Fix $N \geq 3$ and write $x = (x_0, \ldots, x_{N-1}) \in \mathbb{R}^N$.

Definition 3 (Labelled caterpillar, `Lean_3`)**.** A *caterpillar description* is a permutation–set pair

$$\pi \in \mathrm{Sym}\{0, \ldots, N-1\}, \qquad \varnothing \neq S \subseteq \{1, \ldots, r\}, \qquad r := N - 2. \tag{9}$$

Position j carries label $\pi(j+1)$, and $\pi(0)$ is the *apex*. The *spine positions* are $\{0\} \cup S$. The *cut set* S separates positions $j-1$ and j whenever $j \in S$. The last spine position q and the *parent* $p(j)$, the nearest preceding spine position, are

$$q := \max S, \qquad p(j) := \max\big(\{0\} \cup \{i \in S : i < j\}\big) \qquad (1 \leq j \leq r). \tag{10}$$

The parent–child edges form the *labelled caterpillar*: a path through the spine positions with each remaining label attached as a leaf to its parent (Figure 1). The apex is separate from the tree.

Definition 4 (Caterpillar function and face functions, `Lean_4`, `Lean_5`)**.** For labels i, j, for the tree edges, and for the head, put

$$\ell_{ij}(x) := x_i - x_j, \tag{11}$$

$$L_{\pi,S} := \{\ell_{\pi(p(j)+1),\pi(j+1)} : 1 \le j \le r\}, \tag{12}$$

$$h_{\pi,S} := \ell_{\pi(q+1),\pi(0)}. \tag{13}$$

The *edge functions* $L_{\pi,S}$ subtract each child's coordinate from its parent's, one for each of the r tree edges, and the *head function* $h_{\pi,S}$ compares the last spine vertex with the apex. For a linear function h, a finite set L of linear functions, and a *state* $\tau(\ell) \in \{0,\ \ell,\ \sigma \circ \ell\}$ chosen for every $\ell \in L$, define

$$U^{\tau}_{h,L}(x) := \sigma\Big(h(x) + \sum_{\ell \in L} \tau(\ell)(x)\Big). \tag{14}$$

Each edge is omitted, kept linearly, or kept as a hinge. Write $U_{h,L}$ for the all-hinge state, and call (h, L) its *function description*, recording the head and inner hinges. The *caterpillar function* of (π, S) and its *face functions* are

$$U_{\pi,S} := U_{h_{\pi,S},L_{\pi,S}}, \qquad U^{\tau}_{\pi,S} := U^{\tau}_{h_{\pi,S},L_{\pi,S}}. \tag{15}$$

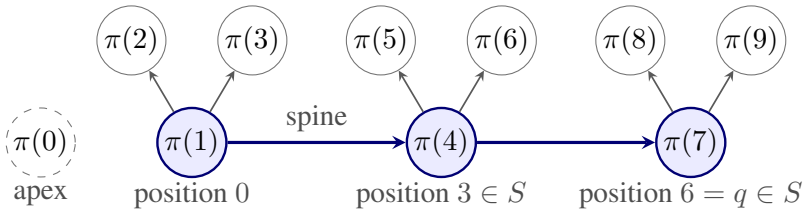


Figure 1: A labelled caterpillar with $N = 10$ and $S = \{3, 6\}$. The spine $\pi(1) \to \pi(4) \to \pi(7)$ has two leaves at each vertex; all arrows point from parent to child. The apex $\pi(0)$ is separate from the tree. The last spine position is $q = 6$, so the head function is $h_{\pi,S}(x) = x_{\pi(7)} - x_{\pi(0)}$.

The term *face function* comes from the following geometry. Identify $a(x) = \langle a, x\rangle$ with its coefficient vector. Then $\ell_{ij} = e_i - e_j$ points from e_j to e_i, and $h = h_{\pi,S} = e_{\pi(q+1)} - e_{\pi(0)}$ points from the apex label to the last spine label. Each $\ell \in L = L_{\pi,S}$ points from child to parent. As free vectors, h places a base corner and the ℓ generate its edges. The polytope apex is 0, distinct from the labelled point $e_{\pi(0)}$. Define the *support function* and the caterpillar polytope by

$$h_P(x) := \max_{p \in P} \langle p, x\rangle, \tag{16}$$

$$Q_{h,L} := \operatorname{conv}\Big(\{0\} \cup \Big(h + \sum_{\ell \in L} [0, \ell]\Big)\Big), \qquad U_{h,L} = h_{Q_{h,L}}. \tag{17}$$

Indeed, the segment $[0, \ell]$ has support function $\sigma \circ \ell$, Minkowski sums add support functions, and adjoining 0 applies the outer ReLU. The base is the translated zonotope $h + \sum_{\ell \in L}[0, \ell]$. For a state τ in Equation (14), set $t_\ell = \mathbb{1}_{\{\tau(\ell) = \ell\}}$, where $\mathbb{1}$ denotes the indicator. It is zero for omitted edges and hinges; the latter contribute segments. The corresponding polytope is $Q^{\tau}_{h,L} := \operatorname{conv}\big(\{0\} \cup (h + \sum_{\ell \in L} t_\ell \ell + \sum_{\tau(\ell) = \sigma \circ \ell}[0, \ell])\big)$, and $U^{\tau}_{h,L} = h_{Q^{\tau}_{h,L}}$. Figure 3 illustrates the translated generators for $N = 4$ on the slice $x_0 = 0$, where a becomes (a_1, a_2, a_3) and e_0 is represented by 0.

The head connects the apex label to the tree, so the head and $N-2$ edge functions are linearly independent. Thus the base is an $(N-2)$-dimensional zonotope with affine hull avoiding 0, and $Q_{h,L}$ is an $(N-1)$-dimensional pyramid. Each nontrivial face containing 0 cones over a base face. Maximizing on each base segment selects 0, ℓ, or $[0, \ell]$, with support functions 0, ℓ, and $\sigma \circ \ell$; the outer ReLU adjoins the apex. These are precisely the three states in Equation (14).

Definition 5 (Dictionary, `Lean_6`)**.** The *dictionary* $\mathcal{D}$ is the finite set of functions $\mathbb{R}^N \to \mathbb{R}$ consisting of every face function $U^\tau_{\pi,S}$, over all caterpillar descriptions (π, S) and all states τ, together with the *auxiliary functions*

$$\begin{aligned} &x \mapsto \sigma(x_i), \quad x \mapsto \sigma(-x_i) \quad (0 \le i < N), \\ &x \mapsto \sigma(x_i - x_j) \qquad\qquad\quad (0 \le i < j < N). \end{aligned} \tag{18}$$

Heads and edge functions are coordinate differences, so face functions and the functions $x \mapsto \sigma(x_i - x_j)$ are translation invariant: with $\mathbf{1} := (1, \dots, 1) \in \mathbb{R}^N$,

$$U^\tau_{\pi,S}(x + t\mathbf{1}) = U^\tau_{\pi,S}(x) \qquad (t \in \mathbb{R}). \tag{19}$$

Only $\sigma(\pm x_i)$ change under this translation. Theorem 13 proves that $\mathcal{D}$ spans the maximum. Section 7 chooses rational coefficients from one basis of its span, without integration.

3.1 The network

The network computes an expansion of $\max_N$ in face functions and auxiliary functions. To compile face functions, first absorb every linear edge of a face state into the head:

$$U^\tau_{h,L} = U_{h',M}, \qquad h' := h + \sum_{\tau(\ell)=\ell} \ell, \qquad M := \{\ell \in L : \tau(\ell) = \sigma \circ \ell\}. \tag{20}$$

By Equation (8), reversing each remaining ℓ_{ij} with $i > j$ gives the same function with the *normalized description*

$$\begin{aligned} M^+ &:= \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in M\}, \\ h^+ &:= h' + \sum_{\ell_{ij} \in M,\ i>j} \ell_{ij}. \end{aligned} \tag{21}$$

Geometrically, $[0, \ell] = \ell + [0, -\ell]$: reversing a generator shifts the head by the old generator. Normalization chooses a consistent orientation and a corresponding base corner; it does not rescale vectors or change the polytope (Figure 3). Write $U_k := U_{h,M}$ for a normalized description $k = (h, M)$, and let $\mathcal{K}_0$ be the set of normalized descriptions of all face functions. Let A_N be the span of the auxiliary functions Equation (18). The network has three parts (Figure 2).

- The *first hidden layer* computes the shared features

$$\Phi(x) := \big(\sigma(x_0),\ \sigma(-x_0),\ (\sigma(x_i - x_0))_{i>0},\ (\sigma(x_i - x_j))_{i<j}\big). \tag{22}$$

- The *second hidden layer* computes the retained units U_k, $k \in K$, for a set $K \subseteq \mathcal{K}_0$, and the two neurons $\sigma(R)$ and $\sigma(-R)$ for an auxiliary remainder $R \in A_N$.
- The *output layer* returns $\sum_{k \in K} d_k U_k + \sigma(R) - \sigma(-R)$.

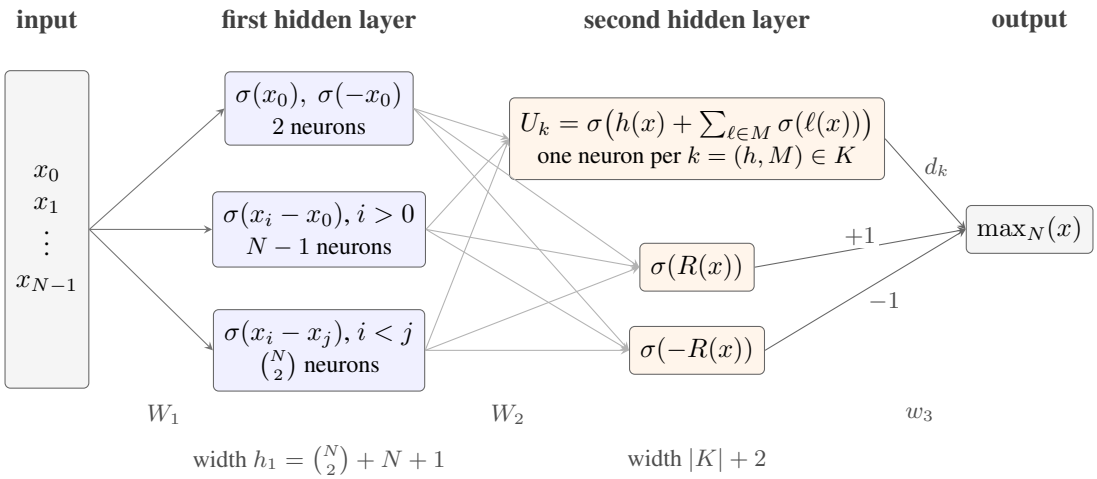


Figure 2: The Caterpillar Net. The first hidden layer is fixed. The second hidden layer has one neuron for every retained normalized description $k = (h, M)$ and two neurons for the auxiliary remainder R. All biases are zero; the matrices are given in Equation (29).

Theorem 6 (Compilation, `Lean_7`, Proof)**.** *Let $K \subseteq \mathcal{K}_0$, $d \in \mathbb{R}^K$, and $R \in A_N$ satisfy*

$$\max_N = \sum_{k \in K} d_k U_k + R. \tag{23}$$

Then the network with first hidden layer Equation (22) *and the matrices in Equation* (29) *computes* $\max_N$*. It has widths h_1 and $|K| + 2$ and zero biases. Its weights are rational whenever the coefficients d_k and the auxiliary coefficients of R are rational.*

Proof of Theorem 6. The first hidden layer outputs Φ from Equation (22). Thus W_1 has rows $e_0^\top$ and $-e_0^\top$, then $e_i^\top - e_0^\top$ for $i > 0$, then $e_i^\top - e_j^\top$ for $i < j$, using a fixed ordering within each block. Its row count is

$$\binom{N}{2} + (N-1) + 2 = \binom{N}{2} + N + 1 = h_1. \tag{24}$$

By Equation (8), these features recover all coordinates,

$$\begin{aligned} x_0 &= \sigma(x_0) - \sigma(-x_0), \\ x_i &= x_0 + \sigma(x_i - x_0) - \sigma(x_0 - x_i) \qquad (i > 0). \end{aligned} \tag{25}$$

For each $k = (h, M) \in K$, substitute Equation (25) into $h(x) + \sum_{\ell \in M} \sigma(\ell(x))$ and collect the coefficients of $\Phi(x)$ into a row vector $w_k^\top$. Every hinge in M already has the orientation $i < j$, so its coefficient is added directly to the corresponding feature.

To specify the remaining row, collect the auxiliary coefficients of R as

$$R(x) = \sum_i p_i \sigma(x_i) + \sum_i n_i \sigma(-x_i) + \sum_{i<j} s_{ij} \sigma(x_i - x_j). \tag{26}$$

Each retained unit is translation invariant by Equation (19), and $\max_N(x + t\mathbf{1}) = \max_N(x) + t$, so Equation (23) gives

$$R(x + t\mathbf{1}) = R(x) + t, \qquad \sum_i p_i = 1. \tag{27}$$

The coefficient sum follows by setting $x = 0$ and $t = 1$. Now choose t large enough that every $x_i + t$ is nonnegative. Substituting the auxiliary expansion into Equation (27) cancels t and gives

$$R(x) = \sum_i p_i x_i + \sum_{i<j} s_{ij}\sigma(x_i - x_j). \tag{28}$$

Substitute Equation (25) into this expression and collect the coefficients of $\Phi(x)$ into w_R. Thus both w_k and w_R are specified by finite sums of the previously defined coefficients. The entries of W_1 and of the rows w_k are integers, because heads and hinges are integer combinations of coordinates. The row w_R is a rational combination of the p_i and s_{ij}, and w_3 consists of the d_k and ± 1.

Order K and set the second hidden layer matrix and output layer vector to

$$W_2 = \begin{pmatrix} (w_k^\top)_{k\in K} \\ w_R^\top \\ -w_R^\top \end{pmatrix}, \qquad w_3 = \begin{pmatrix} (d_k)_{k\in K} \\ 1 \\ -1 \end{pmatrix}, \qquad b_1 = b_2 = b_3 = 0. \tag{29}$$

The second hidden layer outputs $U_k(x)$ and $\sigma(\pm R(x))$. By Equation (8) and Equation (23),

$$w_3^\top \sigma(W_2\Phi(x)) = \sum_{k\in K} d_k U_k(x) + \sigma(R(x)) - \sigma(-R(x)) = \max_N(x). \tag{30}$$

□

Theorem 14 gives such an expansion with rational coefficients; Section 6 counts the normalized descriptions that can occur.

4 Lower bounds for caterpillar expansions

The second width of the network of Theorem 6 is the number of caterpillar functions in an expansion of $\max_N$, plus two. This section shows that this number must grow exponentially, and that the vocabulary itself is large.

Definition 7 (Caterpillar expansion, `Lean_8`)**.** A finite set $K \subseteq \mathcal{K}_0$ is a *caterpillar expansion* of $\max_N$ if there are real numbers d_k, $k \in K$, and a function $R \in A_N$ with $\max_N = \sum_{k\in K} d_k U_k + R$ on $\mathbb{R}^N$. Its *size* is $|K|$.

Theorem 8 (Exponential count, `Lean_9`, `Lean_10`, `Lean_11`, Proof)**.** *Let $N \geq 3$ and let K be a caterpillar expansion of $\max_N$. Then*

$$\binom{N}{m} \leq 6^m\,|K| \qquad (3 \leq m \leq N), \tag{31}$$

$$7^N \leq (N+1)\,6^N\,|K|, \tag{32}$$

$$N(N-1) \leq 6\,|K|. \tag{33}$$

In particular $|K| \geq (7/6)^N/(N+1)$, and every network of the form of Theorem 6 for $\max_N$ has second width at least $(7/6)^N/(N+1) + 2$.

The bound Equation (32) follows from Equation (31) by summing over all m with the binomial theorem; it exceeds the quadratic bound Equation (33) only for N of order 70. The idea of the proof (Appendix C) is local. At a point where $m \geq 3$ coordinates tie at the top and the others lie strictly below in a fixed order,

the m-th mixed difference in the tied coordinates annihilates every auxiliary function, which depends on at most two coordinates, and does not annihilate $\max_N$. Hence some U_k with $k \in K$ is nonlinear in all tied coordinates there. The caterpillar structure allows a single U_k to do so only at a fraction $6^m/\binom{N}{m}$ of these points.

The same bound applies to a larger vocabulary. For an integer head h and a set M of oriented edges of a forest, write $U(x) = \sigma(\sum_i h_i x_i + \sum_{(i,j)\in M} \sigma(x_i - x_j))$ and $\kappa_i = h_i + |\{j : (i,j) \in M\}|$. Delete, simultaneously, all degree-one vertices with $\kappa_i = 0$ from the forest; call the result its *reduced forest*.

Theorem 9 (Forest expansions, `Lean_12`, Proof)**.** *Let $\Delta \geq 2$ be an integer. Suppose $\max_N = \sum_{k\in K} d_k U_k + R$, where $R \in A_N$ and every U_k has an integer head and a forest of functions $\sigma(x_i - x_j)$ whose reduced forest has maximum degree at most Δ. Put $b = 2\Delta + 2$. Then*

$$\binom{N}{m} \leq b^m |K| \qquad (3 \leq m \leq N). \tag{34}$$

Consequently $|K| \geq (1 + 1/b)^N/(N+1)$ for $N \geq 3b + 2$. For $\Delta = 2$, the latter bound holds for every $N \geq 3$.

Caterpillar functions have reduced degree at most 2: their retained leaf edges have $\kappa_i = 0$ at the leaf, and their remaining edges lie on a path. The rooted stars $\sigma(x_c - x_a + \sum_{l\neq a,c} \sigma(x_c - x_l))$ and all their face functions have reduced degree 0. Thus the bound $(7/6)^N/(N+1)$ counts all functions in any mixture of these families, including the building blocks of the direct combinatorial construction of Brunck et al. [BHR+26]; see Appendix D. More generally, Theorem 9 covers every tree of bounded degree, with arbitrary edge orientations, and permits unboundedly many deleted leaves at each vertex.

Theorem 10 (Paths and cycles with arbitrary weights, Proof)**.** *Suppose $N \geq 3$ and $\max_N = R + \sum_{k\in K} d_k U_k$, where R is a linear function plus a finite linear combination of ReLUs of linear forms, and*

$$U_k(x) = \sigma\left(h_k(x) + \sum_{(i,j)\in M_k} c_{k,ij}\sigma(x_i - x_j)\right). \tag{35}$$

The heads h_k are arbitrary linear functions, all coefficients are arbitrary real numbers, and each M_k orients a simple graph of maximum degree at most two. Then

$$\binom{N}{m} \leq \left(N + \frac{m-1}{2}N^2\right)|K| \qquad (3 \leq m \leq N). \tag{36}$$

Taking $m = \lfloor N/2 \rfloor$ gives $|K| = \Omega(2^N/N^{7/2})$ as $N \to \infty$. The conclusion remains valid when biases are allowed.

The comparison graphs in Theorem 10 may be disconnected unions of paths and cycles. The same locality argument also gives an exponential bound for arbitrary weighted forests of bounded degree.

Theorem 11 (Bounded degree with arbitrary weights, Proof)**.** *Let $N \geq 3$ and $\Delta \geq 0$ be integers. Use the representation and coefficient assumptions of Theorem 10, and put $a_N = 1 + N/2 + \binom{N}{2}/4$. If each comparison graph is a forest of maximum degree at most Δ, then*

$$|K| \geq \frac{1 - a_N(2/3)^N}{a_N}\left(\frac{27}{23}\right)^{N/(\Delta+1)}. \tag{37}$$

For arbitrary comparison graphs of maximum degree at most Δ, the same bound holds with the exponential factor replaced by $(1 - (2/3)^\Delta/3)^{-N/(\Delta^2+1)}$. These bounds also hold for the number of distinct comparison graphs used.

Degree in these two theorems refers to the full graph, before any leaf deletion. Every fixed Δ therefore gives an exponential lower bound, with arbitrary orientations, heads, weights, and biases. The proofs use the rank of the comparisons at a tied maximum and elementary counting; they are not yet formalized in Lean. The exponent in Equation (37) depends on Δ, so this does not settle the unrestricted all-tree bound.

Trees with few out-leaves (`Lean_13`). There is also an exponential bound without restricting internal branching. In the normalized description of a tree face function, call a vertex an *out-leaf* if it has degree one in the retained forest and $\kappa_i = 1$. If every vertex of every tree function is adjacent to at most $s \geq 1$ out-leaves, then any expansion by these functions together with caterpillar functions satisfies $|K| \geq (1 + 1/(2s+4))^N/(N+1)$ for $N \geq 3$, where K counts both families. Thus the relevant restriction concerns the oriented edges and head of each function, rather than just the shape of its original tree.

For example, take *any* tree T on the labels other than an apex a, choose vertices s, r of T, orient every edge away from s, and retain all the functions $\sigma(x_p - x_q)$ on those oriented edges. The resulting function is $U_{a,s,r,T}(x) = \sigma(x_r - x_a + \sum_{p \to q \text{ in } T} \sigma(x_p - x_q))$. At a tree vertex v, normalization gives $\kappa_v = \deg_T(v) - 1 + \mathbf{1}_{v=s} + \mathbf{1}_{v=r}$. Only s and r can therefore be out-leaves. A degree-two covering forest improves the bound in this case to $(7/6)^N/(N+1)$, even when these functions are added to the caterpillar and rooted-star face dictionaries; the argument is given at the end of Appendix C.1. This includes every underlying tree shape. It does not include arbitrary face functions of those added trees: replacing edges by linear terms can create many out-leaves at one vertex.

Arbitrary orientations and large matchings. A different argument removes the orientation restriction completely when the original trees have sufficiently many disjoint edges. Let $N \geq 3$ and let k be an integer with $0 \leq 2k \leq N-1$. Adjoin to the caterpillar and rooted-star face dictionaries all tree face functions whose original $(N-1)$-vertex tree contains a matching of size at least k, meaning k edges with pairwise disjoint endpoints. Use any head $x_r - x_a$, any edge orientations, and any of the three face states. Every expansion of $\max_N$ in this enlarged dictionary, with an auxiliary remainder, uses at least $(169/147)^k/(N+1)^2$ functions. The proof at the end of Appendix C.1 is mathematical; this extension is not yet formalized in Lean.

Thus every orientation and every face state is covered when $k \geq \varepsilon(N-1)$ for a fixed $\varepsilon > 0$. This includes all trees with a perfect or near-perfect matching, and every star whose edges are subdivided once. More generally, a tree on $n \geq 2$ vertices with L leaves has a matching of size at least $(n - L + 1)/2$, so a fixed positive fraction of nonleaf vertices suffices. The enlarged dictionary still spans the maximum because it contains the original caterpillar dictionary.

Stars with arbitrary orientations. All star face functions can also be adjoined, with arbitrary head roots, edge orientations, and face states. The caterpillar dictionary together with this entire star family still requires at least $(7/6)^N/(N+1)^4$ nonstar terms for $N \geq 5$. If the matching class above is included as well, the corresponding bound is $(169/147)^k/(N+1)^4$ on the terms outside the added star family. This follows by propagating an exactly checked five-label separator through coordinate deletions; the argument is given in Appendix C.1. It is not yet a Lean result. Nonstar trees with sublinear matching number remain unresolved in general, so the unrestricted all-tree exponential bound is still open.

Theorem 12 (Vocabulary dimension, `Lean_14`, Proof)**.** *For every $N \geq 3$, the span $V_N := \operatorname{span} \mathcal{D}$ of the dictionary satisfies $q_N := \dim(V_N/A_N) \geq (N-2)!$.*

Thus Theorem 14 cannot give a second width below $(N-2)! + 2$; together with Theorem 1, $\log q_N = \Theta(N \log N)$. These bounds concern the stated dictionaries; they do not give a width lower bound for general two-hidden-layer networks.

5 Why the dictionary spans the maximum

The maximum is the support function of a simplex. To work in its affine dimension, set

$$\Delta := \operatorname{conv}\{0, e_1, \dots, e_{N-1}\} \subseteq H_0 := \{x \in \mathbb{R}^N : x_0 = 0\}. \tag{38}$$

For every $x \in \mathbb{R}^N$,

$$\max_N(x) = x_0 + h_\Delta(x - x_0\mathbf{1}). \tag{39}$$

It suffices to prove that h_Δ lies in the span of the restrictions of $\mathcal{D}$ to H_0: the substitution $x \mapsto x - x_0\mathbf{1}$ fixes every face function, sends coordinate features to $\sigma(\pm(x_i - x_0))$, and preserves the dictionary span by Equation (8).

Henceforth all polytopes lie in H_0, and heads and edge functions are identified with their restricted coefficient vectors. Let $\mathcal{L}_k$ be the real span of support functions of polytopes of affine dimension at most k; thus $\mathcal{L}_0$ consists of linear functions. The *exposed face* of P in direction $u \in H_0$ is

$$P^u := \{p \in P : \langle p, u\rangle = h_P(u)\}. \tag{40}$$

The opposite face P^{-u} consists of the minimizers. Define

$$D_u h_P(x) := h_{P^u}(x) - h_{P^{-u}}(x) \qquad (x \in H_0). \tag{41}$$

Here u selects the faces, while x is the support-function input; $P^0 = P$ gives $D_0 = 0$. For a simplex facet normal, D_u subtracts the opposite vertex's linear support function from that of the facet.

For a finite linear combination g of support functions, define its one-sided directional derivative at u by

$$\begin{aligned} \partial_u g(x) &:= \lim_{t \downarrow 0} \frac{g(u + tx) - g(u)}{t}, \\ \partial_u h_P = h_{P^u}, &\qquad D_u g = \partial_u g - \partial_{-u} g. \end{aligned} \tag{42}$$

This makes D_u independent of the support-function decomposition. When $\langle u, \cdot\rangle$ is constant on P, the exposed faces cancel; otherwise both have smaller dimension. Consequently,

$$D_u \mathcal{L}_k \subseteq \mathcal{L}_{k-1} \qquad (k \geq 1). \tag{43}$$

The signed caterpillar sum. We seek a combination that agrees with h_Δ up to a function in $\mathcal{L}_{N-2}$. Adding $x_{\pi(0)}$ makes the caterpillar function the following maximum

$$x_{\pi(0)} + U_{\pi,S}(x) = \max\left\{ x_{\pi(0)},\, x_{\pi(q+1)} + \sum_{\ell \in L_{\pi,S}} \sigma(\ell(x)) \right\}. \tag{44}$$

Averaging over permutations gives the required symmetry

$$F_N(x) := \frac{1}{N!} \sum_{\pi} \sum_{\varnothing \neq S \subseteq \{1,\dots,r\}} (-1)^{N+|S|} \big(x_{\pi(0)} + U_{\pi,S}(x)\big), \tag{45}$$

The signs cancel the extra exposed-face contributions modulo $\mathcal{L}_{N-3}$, as proved below.

Theorem 13 (Dictionary sufficiency, `Lean_15`, Proof)**.** *For every $N \geq 3$, the dictionary of Definition 5 satisfies*

$$\max_N \in \operatorname{span} \mathcal{D}. \tag{46}$$

Proof of Theorem 13. Define the error by

$$f := F_N|_{H_0} - h_\Delta. \tag{47}$$

Step 13.1 reduces dictionary membership to $f \in \mathcal{L}_{N-2}$; Step 13.2 reduces this to $D_u f \in \mathcal{L}_{N-3}$ for all u. Step 13.3 proves these bounds by cancellation, using induction only when deleting a single exceptional coordinate.

Step 13.1 (Reduce to a lower-dimensional error). Assume temporarily that

$$f = F_N - h_\Delta \in \mathcal{L}_{N-2}. \tag{48}$$

Write $Q_{\pi,S} := Q_{h_{\pi,S},L_{\pi,S}}$. Removing the linear apex terms from Equation (45) gives

$$\frac{1}{N!}\sum_{\pi}\sum_{\varnothing\neq S\subseteq\{1,\dots,N-2\}}(-1)^{N+|S|}h_{Q_{\pi,S}} - h_\Delta \in \mathcal{L}_{N-2}. \tag{49}$$

We correct the error by subtracting derivatives ∂_u until it is linear. Faces of $Q_{\pi,S}$ already have support functions in the dictionary span: apex–base faces give Equation (14), base faces give linear functions plus hinges, and the apex alone gives zero. Coordinate features and Equation (8) represent the linear terms and either hinge orientation. Differentiating h_Δ also produces simplex faces; induction on their vertex count supplies these functions.

Claim 13.1 [Simplex faces, `Lean_16`, `Lean_17`, Proof]. For every nonempty $T \subseteq \{0, e_1, \dots, e_{N-1}\}$, the support function of $\Delta_T := \operatorname{conv} T$ lies in the dictionary span.

Start ($|T| = 1$). The support function of a singleton is linear and lies in the dictionary span.

Step ($|T| \geq 2$). Fix T and assume the claim for all smaller nonempty vertex sets. For any current vertex set A with $|A| \geq 2$, remove $v \in A$ using a direction whose value at v exceeds its common value at the other vertices. Writing $\Delta_A := \operatorname{conv} A$, this gives

$$\begin{aligned} u_v &:= \begin{cases} -\sum_{i=1}^{N-1} e_i & v = 0, \\ v & v \neq 0, \end{cases} \\ (\Delta_A)^{u_v} &= \{v\}, \qquad (\Delta_A)^{-u_v} = \Delta_{A\setminus\{v\}}, \\ D_{u_v} h_{\Delta_A}(x) &= \langle v, x\rangle - h_{\Delta_{A\setminus\{v\}}}(x). \end{aligned} \tag{50}$$

Order the vertices outside T as $v_1, \dots, v_{N-|T|}$ and set

$$\varepsilon_T := D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} \left(\frac{1}{N!}\sum_{\pi}\sum_{\varnothing\neq S\subseteq\{1,\dots,N-2\}}(-1)^{N+|S|}h_{Q_{\pi,S}} - h_\Delta \right). \tag{51}$$

Each deletion changes the sign of the remaining simplex term, and subsequent D_u annihilate the linear terms. Hence

$$D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} h_\Delta = (-1)^{N-|T|} h_{\Delta_T} + \ell_T, \tag{52}$$

where ℓ_T is linear. Thus the coefficient of h_{Δ_T} in ε_T is $(-1)^{N-|T|+1}$; all other terms are linear or caterpillar-face support functions and lie in $\operatorname{span} \mathcal{D}$. By Equation (43), $\varepsilon_T \in \mathcal{L}_{|T|-2}$. For the full vertex set the empty composition is the identity.

Lower the residual dimension recursively by

$$\begin{aligned} \varepsilon_T^{(0)}(x) &:= \varepsilon_T(x), \\ \varepsilon_T^{(j+1)}(x) &:= \varepsilon_T^{(j)}(x) - \sum_{u \in E_{T,j}} \big(\partial_u \varepsilon_T^{(j)}\big)(x). \end{aligned} \tag{53}$$

Here $0 \le j < |T| - 2$; the next claim chooses each finite $E_{T,j} \subseteq H_0$.

Claim 13.2 [Residual reduction, `Lean_18`, Proof]. The sets $E_{T,j}$ can be chosen successively so that, for $0 \le j \le |T| - 2$,

$$\begin{aligned} \varepsilon_T^{(j)} &\in \mathcal{L}_{|T|-2-j}, \\ \varepsilon_T - \varepsilon_T^{(j)} &\in \operatorname{span} \mathcal{D}. \end{aligned} \tag{54}$$

Each residual remains a linear combination of h_{Δ_T}, support functions of its proper faces and of caterpillar faces, and linear functions.

The proof in Appendix A.1 chooses a direction orthogonal to each maximal-dimensional direction space, avoiding orthogonality to the simplex and other spaces. The derivative sum preserves each maximal term exactly once; subtracting it lowers the dimension.

At stage $j = |T| - 2$, the error is linear, so Equation (54) gives $\varepsilon_T \in \operatorname{span} \mathcal{D}$. Its simplex coefficient is ± 1, and its other terms already lie in the span; hence $h_{\Delta_T} \in \operatorname{span} \mathcal{D}$. Taking the full vertex set proves $h_\Delta \in \operatorname{span} \mathcal{D}$ under assumption Equation (48).

Step 13.2 (Why the facet bounds imply the class identity). For $N \ge 4$ and $f = F_N - h_\Delta$, we show that $f \in \mathcal{L}_{N-2}$ follows from

$$D_u f \in \mathcal{L}_{N-3} \qquad \text{for every } u \in H_0. \tag{55}$$

The base zonotope admits the Freudenthal–Kuhn triangulation [Fre42, Kuh60]; coning it to the apex expresses each caterpillar support function as a sum of simplex support functions modulo $\mathcal{L}_{N-2}$. For a simplex, D_u isolates a facet up to the linear support function of its opposite vertex. Thus Equation (55) controls the weighted sums of parallel facets. A common linear reconstruction operator turns these facet cancellations back into a cancellation of the original simplex sum. Define

$$(\mathcal{T} g)(x) := \begin{cases} g(x) & x_1 \ge 0, \\ g(x - x_1 e_1) & x_1 < 0. \end{cases} \tag{56}$$

Claim 13.3 [Simplex reconstruction, `Lean_19`, Proof]. Every $(N-1)$-dimensional simplex $P \subseteq H_0$ satisfies

$$h_P - \mathcal{T}\left(\sum_{F \text{ facet of } P} \operatorname{sgn}\langle \nu_F, e_1 \rangle \, h_F \right) \in \mathcal{L}_{N-2}. \tag{57}$$

Here ν_F is an outward normal of F. Its sign against e_1 records whether the facet bounds lines parallel to e_1 from above or below; parallel facets contribute zero. The operator $\mathcal{T}$ fills the segments between a polytope and its projection onto $p_1 = 0$, up to a segment correction for translation; hence $\mathcal{T}\mathcal{L}_k \subseteq \mathcal{L}_{k+1}$. Applying the claim with the same operator to every simplex preserves the cancellations between parallel facets. Their sums lie in $\mathcal{L}_{N-3}$, so their reconstructions lie in $\mathcal{L}_{N-2}$, as required. Appendix A.2 proves the claim and the subdivision and grouping steps.

Claim 13.4 [Class identity, `Lean_20`, Proof]. The class identity Equation (48) holds for every $N \geq 3$.

Step 13.3 (Prove the class identity by induction).

Start ($N = 3$). Here $S = \{1\}$, $q = 1$, and $p(1) = 0$, so Equation (44) gives

$$\begin{aligned} x_{\pi(0)} + U_{\pi,\{1\}}(x) &= \max\bigl\{x_{\pi(0)}, x_{\pi(2)} + \sigma(x_{\pi(1)} - x_{\pi(2)})\bigr\} \\ &= \max\{x_{\pi(0)}, x_{\pi(1)}, x_{\pi(2)}\} = \max_3(x). \end{aligned} \tag{58}$$

All six terms have coefficient $1/6$, so

$$F_3 = \max_3 . \tag{59}$$

On H_0, this gives $f = 0 \in \mathcal{L}_1$.

Step ($N \geq 4$). Assume the class identity for $N - 1$. By Step 13.2, it suffices to prove $D_u f \in \mathcal{L}_{N-3}$ for every $u \in H_0$. The linear apex terms disappear. The exposed simplex faces have the largest and smallest coordinate values u_i, including $u_0 = 0$. In a caterpillar base, $[0, \ell_{ij}]$ remains free when $u_i = u_j$; its maximizing endpoint is ℓ_{ij} when $u_i > u_j$ and 0 when $u_i < u_j$. Minimization reverses these choices.

Which faces can survive? Fix $u \in H_0 \setminus \{0\}$; the direction 0 gives $D_0 f = 0$. A base face is a zonotope, whose support function is linear plus a sum of segment support functions, hence belongs to $\mathcal{L}_1 \subseteq \mathcal{L}_{N-3}$. The apex alone contributes zero. If j tree edges have unequal endpoint values under u, an apex–base face has at most $N - 2 - j$ free segments and therefore dimension at most $N - 1 - j$. Thus only $j \leq 1$ needs attention.

Write $a = u_{\pi(0)}$. An apex–base face requires the maximum over the base to be zero. For $j = 0$, all tree values equal c, so this condition is $c - a = 0$, forcing $u = 0$. For $j = 1$, deleting the unequal edge leaves constant values $c \neq d$ on its two components. The zero-maximum condition is

$$a = \max\{c, d\} \qquad \text{for a spine edge,} \tag{60}$$

$$a = c + \sigma(c - d) \qquad \text{for a leaf edge, with leaf value } d. \tag{61}$$

The spine case gives two coordinate values. The leaf case does too when $d > c$; when $d < c$, it gives the values $c + t, c - t, c, \ldots, c$, where $t = c - d > 0$. Replacing u by $-u$ treats the opposite face. Consequently, only two-value directions and these symmetric three-value directions can contribute modulo $\mathcal{L}_{N-3}$.

Three values: reflection symmetry. Suppose $u_\alpha = c + t$, $u_\beta = c - t$, and all other coordinates equal c. Let τ exchange α and β. On H_0, its coordinate action is

$$(Rx)_i = x_{\tau(i)} - x_{\tau(0)}, \qquad Rx = x - \lambda(x)u, \qquad \lambda(x) := \frac{x_\alpha - x_\beta}{t}. \tag{62}$$

In particular $Ru = -u$. For $G = F_N|_{H_0}$, permutation symmetry and a common coordinate shift give $G \circ R = G + \ell$ with ℓ linear. Every exposed face in direction v lies in the hyperplane $\langle v, p\rangle = h_P(v)$; hence $\partial_v G(x + sv) = \partial_v G(x) + sG(v)$. Applying this to the derivative of $G \circ R = G + \ell$ at u gives

$$\partial_u G(x) + \ell(x) = \partial_{-u} G(Rx) = \partial_{-u} G(x) + \lambda(x) G(-u). \tag{63}$$

Therefore

$$D_u G = \lambda\, G(-u) - \ell \in \mathcal{L}_0. \tag{64}$$

The two exposed simplex faces are vertices, so $D_u f \in \mathcal{L}_0$ as well.

Handling two distinct values. First suppose that each value occurs at least twice. After the preceding dimension bound, every term still requiring cancellation has a unique split $u_{\pi(1)} = \cdots = u_{\pi(k)} \neq u_{\pi(k+1)} = \cdots = u_{\pi(r+1)}$ with $k \in S$. If the apex label has the latter value, toggle membership of $k-1$ in S. Otherwise, exchange the equal-valued labels $\pi(k+1), \pi(k+2)$ and toggle membership of $k+1$. Repetition of both values gives $k \geq 2$ in the first case and $k+2 \leq r+1$ in the second, so the operations are defined; the unchanged cut at k keeps the cut set nonempty. The transformed term remains in the same case, and repeating the operation returns the original term. If Φ denotes this involution on the active terms and $w_{\pi,S} := (-1)^{N+|S|}/N!$, then

$$w_{\Phi(\pi,S)} = -w_{\pi,S}, \qquad D_u h_{Q_{\pi,S}} - D_u h_{Q_{\Phi(\pi,S)}} \in \mathcal{L}_1. \tag{65}$$

Inactive fixed points of the global transformation are already in $\mathcal{L}_{N-3}$ by the dimension bound. Summing the active pairs therefore puts $D_u F_N$ in $\mathcal{L}_{N-3}$. Each exposed simplex face has at most $N-2$ vertices and hence dimension at most $N-3$, so $D_u f \in \mathcal{L}_{N-3}$ without using induction.

One exceptional coordinate: the induction hypothesis. Suppose one of the two values occurs only once. Choose τ with $\tau(0)$ the exceptional label. To stay in H_0, relabel and subtract the new zeroth coordinate:

$$B_\tau : H_0 \longrightarrow H_0, \qquad (B_\tau x)_i := x_{\tau(i)} - x_{\tau(0)} \qquad (0 \leq i < N). \tag{66}$$

Its inverse is $B_{\tau^{-1}}$, and it sends u to $(0, t, \ldots, t)$, $t \neq 0$. Transforming both input and direction preserves every $\mathcal{L}_k$ and changes f only by a linear term, annihilated by D. Rescaling, and reversing the sign of D_u if needed, reduces to $u = (0, 1, \ldots, 1)$.

This direction exposes the $(N-2)$-dimensional facet $\Delta' := \operatorname{conv}\{e_1, \ldots, e_{N-1}\}$ and the opposite vertex 0. Hence

$$D_u h_\Delta(x) = h_{\Delta'}(x) = \max\{x_1, \ldots, x_{N-1}\} \qquad (x \in H_0). \tag{67}$$

The induction hypothesis also applies to unrestricted lower-arity inputs modulo linear functions. Indeed, put $z = (x_1, \ldots, x_{N-1})$ and $y = z - x_1 \mathbf{1}$. Each summand of F_{N-1} increases by the common shift, so

$$F_{N-1}(z) - \max_{N-1}(z) \equiv F_{N-1}(y) - \max_{N-1}(y) \pmod{\mathcal{L}_0}. \tag{68}$$

The first coordinate of y is zero. Induction and linear precomposition therefore give

$$\big(x \mapsto F_{N-1}(x_1, \ldots, x_{N-1}) - \max\{x_1, \ldots, x_{N-1}\}\big) \in \mathcal{L}_{N-3}. \tag{69}$$

Claim 13.5 [Coordinate deletion, `Lean_21`, Proof]. For $N \geq 4$,

$$D_{(0,1,\ldots,1)}(F_N|_{H_0}) - \big(x \mapsto F_{N-1}(x_1, \ldots, x_{N-1})\big) \in \mathcal{L}_{N-3}. \tag{70}$$

The proof is in Appendix A.3. Its mechanism is that only degree-one tree positions of label 0 survive. Each nonempty lower cut set has N insertion preimages, each contributing $1/N$ of its lower coefficient. The extra empty-cut star sum lies in $\mathcal{L}_{N-3}$ by Boolean cancellation.

Completion of the induction. Combining Equation (67), Equation (69), and Claim 13.5 gives

$$\begin{aligned} D_u f(x) = &\big[D_u(F_N|_{H_0})(x) - F_{N-1}(x_1,\dots,x_{N-1})\big] \\ &+ \big[F_{N-1}(x_1,\dots,x_{N-1}) - \max\{x_1,\dots,x_{N-1}\}\big]. \end{aligned} \tag{71}$$

Both brackets lie in $\mathcal{L}_{N-3}$, the first by Claim 13.5 and the second by induction. Step 13.2 now gives $f \in \mathcal{L}_{N-2}$, and Step 13.1 gives the dictionary representation. □

6 Network size

The widths in Theorem 1 are worst-case upper bounds: we count all possible descriptions after collecting linear terms into the head and orienting the remaining hinges consistently. Discarding zero aggregate coefficients can give a smaller network; zero padding realizes the stated widths.

The normalized description Equation (21) of a caterpillar function depends only on its undirected spine, apex, and leaf attachments. For m spine vertices these choices give the summand of C_N in Equation (6). Each of the $N-2$ generators has three face states, giving at most $3^{N-2}C_N$ descriptions. The $N(N-1)$ descriptions with no retained hinge and head $x_i - x_j$, $i \neq j$, can be combined with the auxiliary functions. By Theorem 6, the second width is the number of retained descriptions plus two; Appendix B gives the complete count.

Formalization (`Lean_22`). In this working draft, the theorem and equation links point to the local Lean sources. Appendix F prints the network definitions and the main theorem, and describes the verification certificate.

7 Compression

Theorem 6 reduces the second width to the number of retained units in an expansion. This section bounds that number through the dimension of the vocabulary. Let $V_N := \operatorname{span} \mathcal{D}$, and let $A_N \subseteq V_N$ be the span of the auxiliary functions Equation (18). Put

$$q_N := \dim\big(V_N/A_N\big) = \dim V_N - \dim A_N. \tag{72}$$

Theorem 14 (Quotient compression, `Lean_23`, Proof)**.** *For every $N \geq 3$ there are a set $K \subseteq \mathcal{K}_0$ with $|K| \leq q_N$, rational coefficients d_k, and $R \in A_N$ with rational auxiliary coefficients such that $\max_N = \sum_{k\in K} d_k U_k + R$. Hence $\max_N$ is computed by a network with rational weights, zero biases, and widths*

$$h_1 = \binom{N}{2} + N + 1, \qquad h_2 = q_N + 2. \tag{73}$$

Proof of Theorem 14. Choose a basis of A_N among the auxiliary functions and extend it by face functions to a basis of V_N. The extension adds exactly q_N face functions. For each of them select one normalized description, rather than splitting its coefficient across all descriptions of the same function; this gives K with $|K| \leq q_N$. Let $S = \{g_1,\dots,g_m\}$ be the resulting basis of V_N. By Theorem 13, $\max_N \in V_N$, so $\max_N = \sum_i c_i g_i$ with unique real coordinates c_i.

The coefficients can be made rational without changing the support. Choose a $\mathbb{Q}$-linear map $P : \mathbb{R} \to \mathbb{Q}$ fixing $\mathbb{Q}$ by extending $\{1\}$ to a $\mathbb{Q}$-basis of $\mathbb{R}$. Every g_i and $\max_N$ is continuous and rational-valued on $\mathbb{Q}^N$. For $y \in \mathbb{Q}^N$, applying P to the expansion gives

$$\max_N(y) = P\left(\sum_i c_i g_i(y)\right) = \sum_i P(c_i) g_i(y). \tag{74}$$

Continuity extends the equality to all of $\mathbb{R}^N$, with rational coefficients $P(c_i)$. Theorem 6 compiles the expansion; its weights are rational because the new coefficients are rational. □

Every face function is the unit of a normalized description outside $\mathcal{P}$ or an auxiliary combination (Appendix B), so $q_N + 2$ never exceeds the second width of Theorem 1. The quotient dimension controls only this compression. A network needs only the units of one expansion of $\max_N$; Theorem 15 gives expansions with one and three units for $N = 3, 4$.

Theorem 15 (Small arities, `Lean_24`, `Lean_25`)**.** *For all real inputs,*

$$\max_3(x) = x_1 + \sigma\big(x_2 - x_1 + \sigma(x_0 - x_2)\big), \tag{75}$$

$$\begin{aligned}\max_4(x) = {} & x_1 - \sigma(x_0 - x_2) - \sigma\big(x_1 + x_2 - x_0 - x_3 + \sigma(x_0 - x_1)\big) \\ & + \sigma\big(x_1 + x_2 - x_0 - x_3 + \sigma(x_0 - x_1) + \sigma(x_0 - x_2)\big) \\ & + \sigma\big(x_3 - x_1 + \sigma(x_0 - x_2) + \sigma(x_2 - x_3)\big).\end{aligned} \tag{76}$$

Hence the network of Theorem 6 computes $\max_3$ *with widths* $(7, 3)$ *and* $\max_4$ *with widths* $(11, 5)$*, with rational weights and zero biases.*

Each outer unit in Equations (75) and (76) is a normalized face function. Lean proves both identities by exhaustive case analysis of the activation pattern on all of $\mathbb{R}^N$. For these arities, smaller networks are known: nesting $\max(s, t) = t + \sigma(s - t)$ gives widths $(5, 3)$ for $\max_3$ and $(6, 3)$ for $\max_4$. The identities illustrate how sparse the expansion in the vocabulary can be.

7.1 A matching bound with a fixed first hidden layer

For five inputs, an explicit first hidden layer admits a matching construction and lower bound for the second hidden layer. Define

$$\begin{aligned}\Phi_{10}(x) = \big(&\sigma(x_0 - x_1), \sigma(x_0 - x_2), \sigma(x_1 - x_2), \sigma(x_2 - x_1), \\ &\sigma(x_3 - x_1), \sigma(x_1 - x_4), \sigma(x_4 - x_1), \sigma(x_3 - x_4), \sigma(x_1), \sigma(-x_1)\big).\end{aligned} \tag{77}$$

Theorem 16 (Five inputs with a prescribed first hidden layer)**.** *Among two-hidden-layer networks whose first hidden layer has the ten input weight rows in Equation* (77)*, the minimum second hidden layer width for computing* $\max_5$ *is exactly six. The lower bound allows arbitrary real weights in the remaining layers and arbitrary biases in every layer. The bound is attained with rational weights and zero biases.*

Here is the attaining identity. Put $a = \sigma(x_0 - x_1)$, $b = \sigma(x_2 - x_1)$, $c = \sigma(x_3 - x_1)$, and $d = \sigma(x_4 - x_1)$, and set $\eta = x_4 - x_2 + \sigma(x_3 - x_4) - \sigma(x_0 - x_2)$. Then

$$\begin{aligned}\max_5(x) = {} & x_1 + a + \sigma(\eta) - \sigma\left(\eta + \frac{a + b - c - d}{2}\right) \\ & + \frac{1}{2}\sigma(b - a + d - c) + \frac{1}{2}\sigma(b - a + c - d).\end{aligned} \tag{78}$$

Every displayed preactivation is a linear combination of the features in Equation (77). Writing $x_1 + a = \sigma(x_1 + a) - \sigma(-x_1 - a)$ uses two more neurons, giving widths $(10, 6)$ without skip connections. Appendix E proves the identity directly and gives the reduction and exact finite certificate for optimality. This optimality concerns the prescribed first hidden layer; the unrestricted five-input width frontier is not determined. The finite lower-bound certificate is independently checked with rational arithmetic and is not a Lean theorem.

8 CPWL functions

By the Wang–Sun theorem, every CPWL function on $\mathbb{R}^d$ is a signed sum of maxima of $d+1$ affine functions. These affine functions can be taken from any finite family among which the function selects pointwise, without rescaling them.

Theorem (Koutschan–Moser–Ponomarchuk–Schicho, [KMPS23, Theorem 2.4], `Lean_26`, `Lean_27`)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be continuous, and let $g_1, \dots, g_n : \mathbb{R}^d \to \mathbb{R}$ be affine with $f(x) \in \{g_1(x), \dots, g_n(x)\}$ for every x. Then there are $T \in \mathbb{N}$, signs $\varepsilon_t \in \{-1, 1\}$, and maps $\alpha_t : \{0, \dots, d\} \to \{1, \dots, n\}$ with*

$$f(x) = \sum_{t=1}^{T} \varepsilon_t \max_{d+1}\big(g_{\alpha_t(0)}(x), \dots, g_{\alpha_t(d)}(x)\big) \qquad \text{for every } x \in \mathbb{R}^d. \tag{79}$$

The affine functions of every polyhedral description of a CPWL function satisfy the hypothesis. The Lean proof follows Ovchinnikov's max–min form $f = \max_S \min_{j \in S} g_j$, expands it by inclusion–exclusion into an integer combination of maxima over subsets of $\{g_1, \dots, g_n\}$, and reduces each maximum to subsets of at most $d+1$ functions by an affine dependence among $d+2$ gradients.

Reusing the first hidden layer (`Lean_28`, `Lean_29`). Let $d \geq 2$, and let (h_1, h_2) be the widths of Theorem 1 for $N = d + 1$. Substituting the arguments of each term of Equation (79) into the Caterpillar Net computes f. All terms use only the features $\sigma(g_a - g_b)$ for $a < b$, $\sigma(\pm g_1)$, and $\sigma(g_a - g_1)$, so one first hidden layer of width $\binom{n}{2} + n + 1$ serves every term. Terms with the same argument set $\alpha_t(\{0, \dots, d\})$ compute the same maximum and are merged, and the auxiliary remainders of all terms form one pair $\sigma(\pm R)$. The second hidden layer has width $s(h_2 - 2) + 2$, where s is the number of distinct argument sets and $s \leq \min\big(T, \sum_{m \leq d+1} \binom{n}{m}\big)$. Thus the second hidden layer holds one copy of the Caterpillar Net units for every distinct maximum; only the first hidden layer and the remainder pair are genuinely shared. The second hidden layer and output layer have rational weights and zero biases, and every first hidden layer neuron applies σ to $\pm(g_a - g_b)$ or $\pm g_a$. Hence rational g_i give rational weights and biases, and linear g_i give zero biases.

9 Conclusion

The second width of the Caterpillar Net is the number of caterpillar functions in an expansion of $\max_N$ plus two. For $N = 3, 4$, one and three functions suffice (Theorem 15); for general N, Theorem 14 retains at most q_N of them. Conversely, every such expansion needs at least $(7/6)^N/(N+1)$ caterpillar functions (Theorem 8), so caterpillar building blocks cannot give subexponential second widths. Reducing our upper bound $2^{O(N \log N)}$ to $2^{O(N)}$ remains open. The factorial dimension bound in Theorem 12 limits compression by a basis of the entire dictionary; it does not give a factorial lower bound for representing the single function $\max_N$. Whether the maximum has two-hidden-layer representations of subexponential width with other second-layer functions remains open.

A Supporting proofs for the dictionary theorem

A.1 Residual reduction

We prove the inner induction used in the simplex-face argument of Section 5. Fix a nonempty vertex set T with $|T| \geq 2$, and assume that every proper simplex face on a subset of T is already in the dictionary span, as in the outer induction there.

Proof of Claim 13.2. We argue by induction on j.

Start ($j = 0$). By definition, $\varepsilon_T^{(0)} = \varepsilon_T$. Its dimension bound and support-function expression were established above, and the removed part is zero.

Step ($j < |T| - 2$). Assume the directions for the first j stages have been chosen and the claim holds at stage j. Put $k := |T| - 2 - j \geq 1$. We require each new direction u to be nonorthogonal to the simplex direction space $\operatorname{dir} \Delta_T := \operatorname{span}\{v - w : v, w \in T\}$. Its values on T are then not all equal, so the maximizing vertices form a nonempty proper subset:

$$\begin{aligned} A_u &:= \{v \in T : \langle u, v\rangle = h_{\Delta_T}(u)\}, \qquad \varnothing \neq A_u \subsetneq T, \\ \partial_u h_{\Delta_T} &= h_{\Delta_{A_u}} \in \operatorname{span} \mathcal{D}. \end{aligned} \tag{80}$$

The last membership uses the outer induction hypothesis, since $|A_u| < |T|$. An orthogonal direction would leave h_{Δ_T} unchanged and would not allow this use of the hypothesis.

Choose an auxiliary expression for $\varepsilon_T^{(j)}$ by support functions of polytopes of dimension at most k, and collect their distinct k-dimensional direction spaces. For each such space V, choose $u_V \in V^\perp$ not orthogonal to $\operatorname{dir} \Delta_T$ or to any other collected space V'. This is possible because $\dim \Delta_T = |T| - 1 > k$: each of $\operatorname{dir} \Delta_T$ and V' contains a vector outside V, whose nonzero projection onto $V^\perp$ is not orthogonal to that vector. Adding sufficiently small multiples of these projections avoids each orthogonality condition in turn while preserving those already avoided. Choose $E_{T,j} := \{u_V : V \text{ is a collected direction space}\}$, with $E_{T,j} = \varnothing$ if there are none. In the sum of derivatives over $E_{T,j}$, each term coming from a k-dimensional polytope is preserved exactly once. Every other derivative of that term is the support function of a proper face, while terms of lower dimension remain in $\mathcal{L}_{k-1}$. Subtracting this derivative sum therefore cancels all k-dimensional terms. By the recursive definition Equation (53), the resulting error satisfies $\varepsilon_T^{(j+1)} \in \mathcal{L}_{k-1} = \mathcal{L}_{|T|-3-j}$.

Now compute these same derivatives using the support-function expression in the claim. The derivative of h_{Δ_T} lies in the dictionary span by Equation (80). Differentiating the support function of an already proper simplex face still gives a support function of a proper face of Δ_T, so the same induction hypothesis applies. Differentiating the support function of a face of $Q_{\pi,S}$ gives a support function of a face of the same polytope, whose support functions lie in the dictionary span, and linear functions remain linear. Thus the derivative sum lies in the dictionary span. By Equation (53), $\varepsilon_T - \varepsilon_T^{(j+1)}$ is the sum of the previously removed part and this derivative sum, so it also lies in the dictionary span. Subtracting the derivative sum preserves the stated types of support-function terms. This completes the proof step.

□

A.2 Reconstruction from opposite exposed faces

We justify Step 13.2: for $N \geq 4$, the bounds Equation (55) imply $f \in \mathcal{L}_{N-2}$. We first subdivide the caterpillar polytopes, then prove Claim 13.3, and finally apply it to the resulting weighted simplex sum.

The base of each $Q_{\pi,S}$ is an affine cube. Order its edge vectors by a bijection $\omega : \{1, \dots, N-2\} \to L_{\pi,S}$, add them one at a time, and adjoin the apex:

$$K_{\pi,S,\omega} := \operatorname{conv}\left(\{0\} \cup \left\{h_{\pi,S} + \sum_{a=1}^{k} \omega(a) : 0 \le k \le N-2\right\}\right). \tag{81}$$

These $(N-2)!$ simplices are obtained by coning the *Freudenthal–Kuhn triangulation* of the base to the apex [Fre42, Kuh60]: in cube coordinates, each base simplex occupies one ordering region $1 \ge t_{\omega(1)} \ge \cdots \ge t_{\omega(N-2)} \ge 0$. For a hyperplane cut $P = P_+ \cup P_-$, the support-function identity

$$h_P + h_{P_+ \cap P_-} = h_{P_+} + h_{P_-} \tag{82}$$

shows that subdivision changes the sum of support functions only by support functions of the cutting sections. Hence

$$h_{Q_{\pi,S}} - \sum_{\omega} h_{K_{\pi,S,\omega}} \in \mathcal{L}_{N-2}. \tag{83}$$

Substituting Equation (83) into Equation (45) expresses f as a finite linear combination of support functions of $(N-1)$-dimensional simplices in H_0, plus a remainder in $\mathcal{L}_{N-2}$. The simplex Δ contributes its support function with coefficient -1, and the linear apex terms are included in the remainder.

For a simplex P, an outward facet normal ν_F exposes the facet F on one side and its opposite vertex v_F on the other. Thus

$$D_{\nu_F} h_P(x) = h_F(x) - \langle v_F, x \rangle. \tag{84}$$

Directions exposing no facet on either side give support functions of faces of dimension at most $N-3$. Consequently, Equation (55) says that each weighted, signed sum of parallel-facet support functions belongs to $\mathcal{L}_{N-3}$.

Our assumption controls the weighted facet contributions, whereas our conclusion concerns f. To recover the original sum modulo $\mathcal{L}_{N-2}$, we use the same linear operator for every simplex, so cancellations between facet contributions survive reconstruction. Use the operator $\mathcal{T}$ defined in Equation (56). The sign in this definition concerns the input x. To interpret it geometrically, first suppose that every point $p \in P$ satisfies $p_1 \ge 0$, and write $\operatorname{pr}(p) := p - p_1 e_1$ for projection onto $p_1 = 0$. Since $\langle p, x \rangle - \langle \operatorname{pr}(p), x \rangle = p_1 x_1$, each point dominates its projection when $x_1 \ge 0$, and its projection dominates when $x_1 < 0$. Consequently,

$$\mathcal{T} h_P(x) = \max\{h_P(x), h_{\operatorname{pr} P}(x)\} = h_{\operatorname{conv}(P \cup \operatorname{pr} P)}(x). \tag{85}$$

This convex hull fills the segments with endpoints p and its projection, for $p \in P$, so its dimension is at most $\dim P + 1$. For an arbitrary P, choose $c \ge 0$ such that $P + ce_1$ lies in $p_1 \ge 0$. Translation adds cx_1 to the support function, and $\mathcal{T}(x_1) = \sigma(x_1)$, giving

$$\mathcal{T} h_P(x) = h_{\operatorname{conv}((P+ce_1) \cup \operatorname{pr}(P+ce_1))}(x) - c\sigma(x_1). \tag{86}$$

The correction $\sigma(x_1)$ is the support function of the segment $[0, e_1]$. Thus linearity gives

$$\mathcal{T} \mathcal{L}_k \subseteq \mathcal{L}_{k+1}. \tag{87}$$

Proof of Claim 13.3. Let $k = |\{F \text{ facet of } P : \langle \nu_F, e_1 \rangle \ne 0\}|$ count facets not parallel to e_1. Boundedness ensures both nonzero signs occur, so $k \ge 2$.

Start ($k = 2$). There is one upper facet F^+ and one lower facet F^-. Every line parallel to e_1 that meets P has its two intersection endpoints on these facets. Thus $h_P(x) = h_{F^+}(x)$ when $x_1 \geq 0$, and $h_P(x) = h_{F^-}(x)$ when $x_1 < 0$. When $x_1 \geq 0$, $\mathcal{T}$ evaluates at x, so subtracting $\mathcal{T}(h_{F^+} - h_{F^-})$ from h_P leaves h_{F^-}. When $x_1 < 0$, $\mathcal{T}$ evaluates both facet support functions at $x - x_1 e_1$, where they agree because the facets have the same projection. Their difference is therefore zero, again leaving h_{F^-}. The reconstruction error is therefore $h_{F^-} \in \mathcal{L}_{N-2}$, proving the claim for $k = 2$.

Step ($k > 2$). Assume the claim for $k - 1$, and let P have k facets not parallel to e_1. Two of them have the same nonzero sign of $\langle \nu_F, e_1 \rangle$. Their intersection is an $(N-3)$-dimensional face C. Let $W := \operatorname{span}\{p - q : p, q \in C\}$, and project orthogonally within H_0 onto $W^\perp$. This is a two-dimensional plane, since $\dim H_0 = N - 1$ and $\dim W = N - 3$. All points of C have the same projection c, because their differences lie in W. There are just two vertices of P outside C, so the projected simplex is a triangle with vertices c and the projections of those two vertices. The chosen facets project to its two sides meeting at c.

The chosen facet normals are orthogonal to W, so their signs against e_1 equal their signs against its projection. This projected direction is nonzero, since the chosen facets are not parallel to e_1. If both sides are upper, moving from c opposite to this direction enters the triangle; if both are lower, moving in this direction enters it. Continuing meets the interior of the opposite edge. The inverse image of the line through c in this direction is a hyperplane in H_0 containing C and parallel to e_1. It cuts the edge whose endpoints are the two vertices outside C at an interior point and divides P into two $(N-1)$-dimensional simplices. Each keeps one of the two chosen facets and replaces the other by the new facet parallel to e_1. Its other facets retain their normal directions, so each new simplex has exactly $k - 1$ facets not parallel to e_1.

Write $E(Q)$ for the reconstruction error on the left of Equation (57). Let P_+ and P_- be the two new simplices, let $R := P_+ \cap P_-$ be their common cutting facet, and write F_1, F_2 for the chosen facets and $R_F := R \cap F$ for every other facet of P. With $s_F := \operatorname{sgn}\langle \nu_F, e_1 \rangle$, applying Equation (82) to P and to every split facet gives the exact residual identity

$$E(P) = E(P_+) + E(P_-) - h_R + \mathcal{T}\left(\sum_{\substack{F \text{ facet of } P \\ F \notin \{F_1, F_2\}}} s_F h_{R_F} \right). \tag{88}$$

Indeed, the inherited facets retain their signs, while the two copies of R are parallel to e_1 and therefore have sign zero. By the induction hypothesis, $E(P_+), E(P_-) \in \mathcal{L}_{N-2}$; moreover, $\dim R = N - 2$ and $\dim R_F \leq N - 3$. Equation Equation (87) therefore puts every term on the right of Equation (88) in $\mathcal{L}_{N-2}$. This proves the claim for k and completes the induction.

□

Conclusion for f. Apply Equation (57) to every simplex support-function term in the expansion of f, keeping its original coefficient. A facet F parallel to e_1 has $\langle \nu_F, e_1 \rangle = 0$, so its term in the reconstruction formula is zero and can be omitted. For each remaining occurrence of a facet, choose ν_F outward from its simplex and put

$$u_F := \frac{\nu_F}{\langle \nu_F, e_1 \rangle}, \qquad \langle u_F, e_1 \rangle = 1. \tag{89}$$

Thus the coordinate with index 1 is normalized to 1; the coordinate with index 0 remains zero because $u_F \in H_0$. Oppositely oriented parallel normals give the same u_F. Only the normal is rescaled: the support function and its coefficient are unchanged.

Let $\mathcal{U}$ be the finite set of vectors u_F obtained this way. For each $u \in \mathcal{U}$, define G_u by adding all facet support-function terms with $u_F = u$, retaining their signs and weights. An occurrence of F as a

facet of $K_{\pi,S,\omega}$ contributes $(-1)^{N+|S|}\operatorname{sgn}\langle \nu_F, e_1\rangle\, h_F/N!$, while an occurrence as a facet of Δ contributes $-\operatorname{sgn}\langle \nu_F, e_1\rangle\, h_F$. These are precisely the terms with the same normalized normal, including all repetitions.

More explicitly, define ρ to be the sum of the linear apex terms and the weighted subdivision errors in Equation (83). Then

$$f = \frac{1}{N!}\sum_{\pi}\sum_{\varnothing\neq S\subseteq\{1,\dots,N-2\}}(-1)^{N+|S|}\sum_{\omega} h_{K_{\pi,S,\omega}} - h_\Delta + \rho, \qquad \rho\in\mathcal{L}_{N-2}. \tag{90}$$

For a simplex P, collect its facets with normalized normal u as

$$B_u(P) := \sum_{\substack{F \text{ facet of } P\\ \langle \nu_F, e_1\rangle\neq 0,\ u_F = u}} \operatorname{sgn}\langle \nu_F, e_1\rangle h_F. \tag{91}$$

An empty sum is zero. If $\langle \nu_F, e_1\rangle > 0$, the direction u_F exposes F; if it is negative, $-u_F$ exposes F. The opposite face is a single vertex, so Equation (84) gives the signed facet term up to a linear function. If neither u nor $-u$ exposes a facet of P, both exposed faces have dimension at most $N-3$. Thus $D_u h_P - B_u(P)\in\mathcal{L}_{N-3}$. Applying this to the full weighted expansion, with the same permutations and nonempty cut sets as in Equation (45), gives

$$\begin{aligned} D_u f - G_u = &\sum_{\pi,S\neq\varnothing}\frac{(-1)^{N+|S|}}{N!}\sum_{\omega}\big(D_u h_{K_{\pi,S,\omega}} - B_u(K_{\pi,S,\omega})\big)\\ &- (D_u h_\Delta - B_u(\Delta)) + D_u\rho \in \mathcal{L}_{N-3} \qquad (u\in\mathcal{U}). \end{aligned} \tag{92}$$

Together with Equation (55), relation Equation (92) puts every G_u in $\mathcal{L}_{N-3}$. By linearity, the sum of the reconstructed facet terms is $\mathcal{T}(\sum_{u\in\mathcal{U}} G_u)$, which lies in $\mathcal{L}_{N-2}$ by Equation (87). Its difference from f consists of the reconstruction and subdivision remainders, also in $\mathcal{L}_{N-2}$. Hence $f\in\mathcal{L}_{N-2}$. This proves the implication used in Step 13.2.

A.3 Coordinate deletion

Proof of Claim 13.5. We now determine which terms of the signed sum remain for $u = (0,1,\dots,1)$. Only label 0 has a different coordinate value from the others. Consequently, a tree-edge generator is nonorthogonal to u exactly when one of its endpoint labels is 0.

If label 0 is the apex label, all tree-edge generators are orthogonal to u. The maximizing face is the whole base $B_{\pi,S}$ and the minimizing face is $\{0\}$, so $D_u h_{Q_{\pi,S}} = h_{B_{\pi,S}}\in\mathcal{L}_1\subseteq\mathcal{L}_{N-3}$. If label 0 is a tree vertex of degree at least two, at least two segment generators are nonorthogonal to u. The preceding dimension bound then gives $D_u h_{Q_{\pi,S}}\in\mathcal{L}_{N-3}$ as well. We can therefore discard both kinds of term: it suffices to retain descriptions in which label 0 is a tree vertex of degree one.

To express this condition in the permutation notation, let $p := \pi^{-1}(0)$ be the index of label 0 in the full order $(\pi(0),\dots,\pi(N-1))$. When $p\geq 1$, its tree position is $p-1$. A degree-one vertex is either an endpoint of the spine with no attached leaves or a leaf outside the spine. The retained descriptions are thus exactly

$$\begin{aligned} &p = 1, \quad 1\in S && \text{(root endpoint)},\\ &2\leq p\leq N-1, \quad p-1\notin S && \text{(leaf outside the spine)},\\ &p = N-1, \quad N-2\in S && \text{(last spine endpoint)}. \end{aligned} \tag{93}$$

Indeed, the root has only one child precisely when $1\in S$; a nonroot spine vertex has a parent and also a child unless it occupies the final tree position. Deleting label 0 from the full order now gives an order $\bar\pi$ mapping positions $\{0,\dots,N-2\}$ bijectively to labels $\{1,\dots,N-1\}$.

For $0 \le i \le N-3$, let ι_i insert a gap at the prospective pendant position $i+1$. Given a lower cut set $T \subseteq \{1,\dots,N-3\}$, the three active insertion rules are

$$\begin{aligned} \iota_i(t) &:= \begin{cases} t, & t < i+1, \\ t+1, & t \ge i+1, \end{cases} \\ \text{first end:} \quad & (p,S) = (1, \{1\} \cup \{t+1 : t \in T\}), \\ \text{pendant } i\text{:} \quad & (p,S) = (i+2, \{\iota_i(t) : t \in T\}) \qquad (0 \le i \le N-3), \\ \text{last end:} \quad & (p,S) = (N-1, T \cup \{N-2\}). \end{aligned} \tag{94}$$

In every row the upper permutation is uniquely obtained by inserting label 0 at position p in $\bar\pi$. Conversely, with t restricted to $\{1,\dots,N-3\}$, the deletion rules are $T = \{t : t+1 \in S\}$ for the first end, $T = \{t : \iota_i(t) \in S\}$ for pendant i, and $T = S \setminus \{N-2\}$ for the last end. These operations are inverse. The three cases in Equation (93) correspond respectively to the first-end row, the pendant row with $i = p-2$, and the last-end row. Thus a nonempty T has $N-2$ pendant and two endpoint preimages, whereas $T = \varnothing$ has only the two endpoints because upper cut sets are nonempty.

Extend the lower caterpillar notation to $T = \varnothing$ by setting its last spine position and every parent equal to 0, so its head begins at the root $\bar\pi(1)$. For every lower label order $\bar\pi$ put

$$\begin{aligned} C_{\bar\pi,T}(x) &:= x_{\bar\pi(0)} + U_{\bar\pi,T}(x), \\ q_T &:= \max(\{0\} \cup T), \qquad p_T(j) := \max(\{0\} \cup \{t \in T : t < j\}). \end{aligned} \tag{95}$$

For each upper preimage (π, S) of $(\bar\pi, T)$, the insertion identities and the literal factorial coefficients are

$$\begin{aligned} \text{pendant:} \quad & \frac{(-1)^{N+|S|}}{N!} = -\frac{1}{N}\frac{(-1)^{N-1+|T|}}{(N-1)!}, \quad D_u\big(x_{\pi(0)} + U_{\pi,S}\big) + C_{\bar\pi,T} \in \mathcal{L}_1, \\ \text{endpoint:} \quad & \frac{(-1)^{N+|S|}}{N!} = \frac{1}{N}\frac{(-1)^{N-1+|T|}}{(N-1)!}, \qquad D_u\big(x_{\pi(0)} + U_{\pi,S}\big) - C_{\bar\pi,T} \in \mathcal{L}_1. \end{aligned} \tag{96}$$

Thus every one of the $N-2+2 = N$ preimages contributes $1/N$ of the lower summand modulo $\mathcal{L}_{N-3}$, and together they recover its coefficient in F_{N-1}.

The extra terms are the two endpoint fibres whose deletion leaves $T = \varnothing$, hence a rooted star. By Equation (96), each contributes $1/N$ of the weighted lower summand modulo $\mathcal{L}_1$. For a fixed lower order $\bar\pi$, the nonroot positions are $J = \{0, 2, \dots, N-2\}$, including the original apex position 0. For $a \in J$, let $\bar\pi^{(a)}$ exchange positions 0 and a, keeping the root at position 1; the exchange at $a = 0$ is the identity. Put

$$\begin{aligned} q := |J| = N-2, \qquad & d_a := x_{\bar\pi(1)} - x_{\bar\pi(a)} \quad (a \in J), \\ T_A := \sigma\left(\sum_{a\in A} d_a + \sum_{a \in J\setminus A} \sigma(d_a)\right) \quad & (A \subseteq J), \\ C_{\bar\pi^{(a)},\varnothing} = x_{\bar\pi(a)} + T_{\{a\}} \quad & (a \in J). \end{aligned} \tag{97}$$

The Boolean alternating identity gives

$$\sum_{A\subseteq J} (-1)^{|A|} T_A = 0 \quad \Longrightarrow \quad \sum_{a\in J} T_{\{a\}} = T_\varnothing + \sum_{\substack{A\subseteq J \\ |A|\ge 2}} (-1)^{|A|} T_A. \tag{98}$$

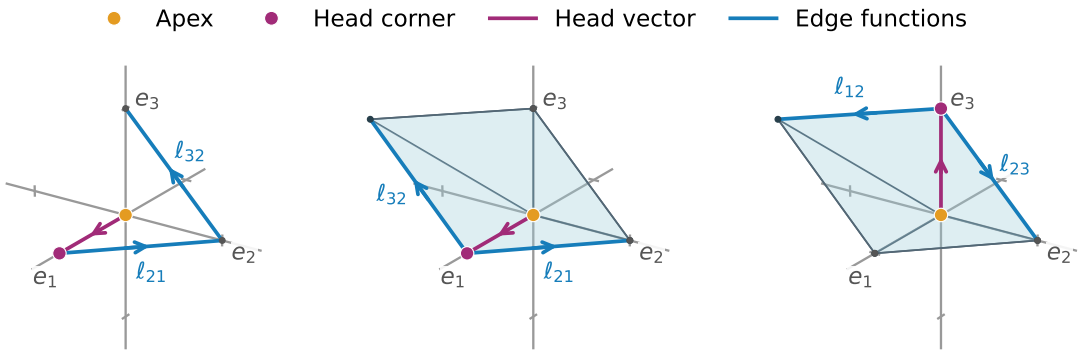
The empty term is a sum of support functions of segments. For $|A| \ge 2$, the polytope with support function T_A has one head direction and $q - |A|$ segment directions, hence dimension at most $1 + q - |A| \le q - 1 = N-3$.

Thus the singleton sum lies in $\mathcal{L}_{N-3}$; the sum of the linear apex coordinates lies in $\mathcal{L}_0$. Each fixed-position exchange is a bijection on lower label orders, so

$$q \sum_{\bar{\pi}} C_{\bar{\pi},\varnothing} = \sum_{\bar{\pi}} \sum_{a \in J} C_{\bar{\pi}^{(a)},\varnothing} \in \mathcal{L}_{q-1}. \tag{99}$$

Since $q > 0$, the full empty-cut sum belongs to $\mathcal{L}_{q-1} = \mathcal{L}_{N-3}$. The empty-cut coefficient $(-1)^{N-1}/(N-1)!$ and the factor $2/N$ from the two endpoint fibres are scalars, so the extra star sum lies there as well. This proves Claim 13.5. □

B Network construction and width bounds



(a) Coordinate differences (b) Original description (c) Normalized description

Figure 3: Coordinate differences and the caterpillar polytope for $\pi = (0, 3, 2, 1)$ and $S = \{1, 2\}$. (a) The head $h = e_1 - e_0$ and edge functions $L = \{\ell_{32} = e_3 - e_2, \ell_{21} = e_2 - e_1\}$ drawn between their labelled endpoints. (b) The same generators translated to the base corner h of $Q_{h,L}$. (c) Normalization (Section 6) gives $h^+ = h + \ell_{32} + \ell_{21} = e_3 - e_0$ and $L^+ = \{\ell_{12} = e_1 - e_2, \ell_{23} = e_2 - e_3\}$, leaving the polytope unchanged. Purple arrows represent the head (the apex–last spine connection before normalization); blue arrows represent the spine edge functions. Orange dots mark the polytope apex and purple dots the chosen head corner. The grey coordinate axes meet at the origin, representing e_0 after restriction to $x_0 = 0$; the positive unit points are labelled e_1, e_2, e_3.

Proof of Theorem 1. Fix $N \geq 3$. We bound the quotient dimension q_N of Section 7 by the count of normalized descriptions and apply Theorem 14.

Step 1.1 (Count the caterpillar face functions). Use the adjusted heads and normalized descriptions from Section 6. Counting distinct normalized descriptions bounds q_N.

For a caterpillar description (π, S), the spine endpoints are labelled $\pi(1)$ and $\pi(q+1)$, and every label other than $\pi(1)$ and the apex is the child of exactly one edge. Hence the normalized description of $U_{\pi,S}$ is

$$\begin{aligned} h^+_{\pi,S}(x) &= \sum_{\ell_{ij} \in L_{\pi,S}} x_{\max(i,j)} + x_{\pi(1)} + x_{\pi(q+1)} - \sum_{i=0}^{N-1} x_i, \\ L^+_{\pi,S} &= \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in L_{\pi,S}\}. \end{aligned} \tag{100}$$

This depends only on the undirected tree, apex, and unordered spine endpoints. For m spine vertices, choose the vertex set in $\binom{N}{m}$ ways, its order up to reversal in $m!/2$ ways, the apex in $N-m$ ways, and each of the $N-m-1$ leaf attachments in m ways. Reversal has no fixed point since $m \geq 2$, so Equation (6) gives

$$\left|\{(h^+_{\pi,S}, L^+_{\pi,S}) : (\pi, S) \text{ a caterpillar description}\}\right| \leq C_N. \tag{101}$$

Reversing an edge swaps its omitted and linear states. Thus choosing states before or after normalization gives the same family $\mathcal{K}_0$ of descriptions, with $|\mathcal{K}_0| \leq 3^{N-2}C_N$. Write $U_k := U_{h,M}$ for $k = (h, M)$. We merge equal descriptions; distinct descriptions may still represent the same function.

Among them are the descriptions $(\ell_{ij}, \varnothing)$ representing $x \mapsto \sigma(x_i - x_j)$, with $i \neq j$: for the all-cut path $S = \{1, \dots, r\}$ with apex $\pi(0) = j$ and last spine vertex $\pi(r+1) = i$, omitting every edge function gives

$$U^0_{\pi,S} = \sigma \circ \ell_{ij} \qquad (0 \leq i \neq j < N). \tag{102}$$

Let $\mathcal{P} \subseteq \mathcal{K}_0$ consist of descriptions with empty hinge set and head a coordinate difference. The distinct heads ℓ_{ij} give $|\mathcal{P}| \geq N(N-1)$. Put

$$K := \mathcal{K}_0 \setminus \mathcal{P}. \tag{103}$$

Every face function in $\mathcal{D}$ equals U_k for its normalized description $k \in \mathcal{K}_0$, and by Equation (8) every unit indexed by $\mathcal{P}$ lies in the auxiliary span A_N. Hence the units U_k, $k \in K$, span V_N modulo A_N, and

$$q_N \leq |K| \leq 3^{N-2}C_N - N(N-1). \tag{104}$$

Step 1.2 (Compile the network and establish its widths). Theorem 14 gives a network with rational weights, zero biases, first width h_1, and second width

$$q_N + 2 \leq 3^{N-2}C_N - N(N-1) + 2 = h_2. \tag{105}$$

Zero padding realizes the stated widths exactly.

For the growth rates, count the ordered choices of m spine labels:

$$\binom{N}{m}\frac{m!}{2} = \frac{N!}{2(N-m)!} \leq N^m \qquad (2 \leq m \leq N-1). \tag{106}$$

Using $N - m \leq N$ and $m^{N-m-1} \leq N^{N-m-1}$ in each summand of Equation (6) gives

$$\begin{aligned} C_N &\leq \sum_{m=2}^{N-1} N^m \, N \, N^{N-m-1} = (N-2)N^N \\ &\leq N^{N+1} \leq N^{4N}. \end{aligned} \tag{107}$$

Since $3^{N-2} \leq N^N$ for $N \geq 3$,

$$h_2 \leq 3^{N-2}(N-2)N^N + 2 \leq 2N^{5N}, \tag{108}$$

$$h_1 \leq 4N^2. \tag{109}$$

Taking logarithms gives the asymptotic bounds in Theorem 1. □

C Proofs of the lower bounds

C.1 Proof of the exponential count bound

We prove Theorem 9 and obtain Equations (31) to (32) by taking $\Delta = 2$. The proof uses one mixed difference and a product over disjoint star blocks.

Proof of Theorem 8. Fix an expansion as in Theorem 9, and $3 \le m \le N$. Write $U_k = \sigma(A_k)$ with $A_k(x) = \sum_o h_{k,o}x_o + \sum_{(i,j)\in M_k} \sigma(x_i - x_j)$. Reorienting edges by Equation (8) preserves A_k and $\kappa_{k,o}$, so we may assume $i < j$ in M_k. Choose an integer $B > 1$ and $\delta > 0$ such that, for every k, $B > \sum_o |h_{k,o}| + |M_k|$ and $\delta(\sum_o |h_{k,o}| + 2|M_k|) < 1$.

Step 8.1 (A mixed difference detects every top tie). A permutation ρ assigns positions to labels, with smaller positions higher. Put $T_\rho = \{o : \rho(o) < m\}$ and define

$$p^\rho_o = \begin{cases} 0, & o \in T_\rho, \\ -B^{\rho(o)}, & o \notin T_\rho, \end{cases} \tag{110}$$

$$\Delta_\rho g = \sum_{S \subseteq T_\rho} (-1)^{m-|S|} g(p^\rho + \delta \mathbf{1}_S). \tag{111}$$

Pairing corners along a missing coordinate shows that Δ_ρ annihilates every function of at most two coordinates. On this cube the maximum is δ at every nonempty corner and 0 at the empty corner. Thus

$$\Delta_\rho R = 0 \qquad (R \in A_N), \tag{112}$$

$$\Delta_\rho \max\nolimits_N = (-1)^{m+1}\delta \neq 0. \tag{113}$$

Some U_k in the expansion therefore satisfies $\Delta_\rho U_k \neq 0$.

Step 8.2 (Detection forces balance). Fix one such unit and suppress its index. Its preactivation at p^ρ is an integer and changes by less than 1 on the cube. Unless $A(p^\rho) = 0$, its outer ReLU is either zero or redundant throughout the cube, and $\Delta_\rho U = 0$. For $o \notin T_\rho$, let $u_\rho(o)$ count its neighbours at smaller positions. Collecting coefficients gives

$$A(p^\rho) = -\sum_{o \notin T_\rho} (\kappa_o - u_\rho(o)) B^{\rho(o)}. \tag{114}$$

The integer digits have absolute value less than B. Successively reducing modulo B forces each digit to vanish. Moreover, if a tied label t has no tied neighbour, the slope in x_t on the cube is κ_t; if this is zero, the mixed difference vanishes. Call U *passing* ρ when these two necessary conditions hold:

$$u_\rho(o) = \kappa_o \qquad \text{for } o \notin T_\rho, \tag{115}$$

$$\kappa_t \neq 0 \qquad \text{for } t \in T_\rho \text{ with no neighbour in } T_\rho. \tag{116}$$

Every permutation is passed by some unit in the expansion.

Step 8.3 (Only the reduced forest chooses the tie set). Fix a unit, let F be its forest, and call the deleted degree-one vertices *pendants*. Let Φ contain the vertices that are neither pendants nor isolated in F. For a passed tie set T, an isolated vertex belongs to T exactly when its κ is nonzero. A pendant belongs to T exactly when its neighbour does: either mismatch contradicts Equations (115) and (116). A component consisting of two pendants must lie entirely in T, since otherwise its lower vertex violates balance. Consequently T is determined by $A = T \cap \Phi$, and $|A| \le m$.

Step 8.4 (Each disjoint edge halves the possible orders). For fixed T, balance prescribes the out-degree of each vertex of $F - T$ when edges point towards the higher endpoint. A forest orientation is uniquely determined by its out-degrees: orient the edge at a leaf, delete the leaf, and repeat. Hence every passed permutation with this T induces the same orientation of $F - T$. If we select a matching of $\mu(A)$ edges of $F - T$, independently swapping their endpoints gives

$$|\{\rho : U \text{ passes } \rho,\ T_\rho = T\}| \le m!\,(N-m)!\,2^{-\mu(A)}. \tag{117}$$

Step 8.5 (Partition into blocks and multiply their weights). Partition Φ into stars of the reduced forest and singletons isolated in that forest. Such a star partition exists: an inclusion-minimal edge cover of the non-isolated vertices contains no three-edge path, since its middle edge could be removed. As it is a forest, its components are stars. Every star has $1 \le s \le \Delta$ leaves. A singleton has a pendant neighbour in F; fix one.

For $A \subseteq \Phi$, select one edge in a star block whenever its centre and some leaf are outside A. At a singleton outside A, select its fixed pendant edge. For a passed tie set these are disjoint edges of $F - T$. Let $\mu(A)$ count them and put $y = 1/(2\Delta + 2) = 1/b$. The sum of $y^{|A|}2^{-\mu(A)}$ factorizes over blocks. A singleton has weight $1/2 + y$; a star with s leaves has weight

$$W_s(y) = y(1+y)^s + \tfrac{1}{2}\big((1+y)^s - y^s\big) + y^s = (\tfrac{1}{2} + y)(1+y)^s + \tfrac{1}{2}y^s. \tag{118}$$

The three terms distinguish a selected centre, an unselected centre with an unselected leaf, and an unselected centre with all leaves selected. All block weights are at most 1. For $\Delta = 2$, the singleton, one-leaf, and two-leaf weights at $y = 1/6$ are respectively $2/3$, $31/36$, and $199/216$. In general $y \le 1/6$; the one-leaf case is bounded by $31/36$, and for $2 \le s \le \Delta$, $W_s(y) \le (1/2 + y)e^{1/2-y} + y^2/2 \le (2/3)e^{1/3} + 1/72 < 1$. Thus

$$\sum_{A \subseteq \Phi} y^{|A|}2^{-\mu(A)} = \prod_{\text{blocks}} W_{\text{block}}(y) \le 1. \tag{119}$$

Step 8.6 (Count all permutations). Distinct passed tie sets have distinct $A = T \cap \Phi$, and $|A| \le m$. Summing Equation (117) and inserting the factor $1 \le b^m y^{|A|}$ bounds the number of permutations passed by one unit by $m!\,(N-m)!\,b^m$ using Equation (119). The units cover all $N!$ permutations, so $N! \le |K|\,m!\,(N-m)!\,b^m$, proving Equation (34). Caterpillar functions have reduced degree at most 2, giving Equation (31).

Step 8.7 (Extract the exponential bound). Put $t_m = \binom{N}{m} b^{-m}$. If $N \ge 3b + 2$, then $t_0 \le t_1 \le t_2 \le t_3$, so every t_m is at most $|K|$. The binomial theorem gives

$$(1 + 1/b)^N = \sum_{m=0}^{N} t_m \le (N+1)|K|. \tag{120}$$

For $b = 6$ and $3 \le N < 20$, the same conclusion follows from $|K| \ge 1$ and $(7/6)^N \le N + 1$. This proves Equation (32) and the remaining assertions of Theorem 9.

Quadratic refinement (`Lean_30`). The bound Equation (33) uses three-element tie sets, a six-point second difference, and the sharper count of at most $N - 2$ tie sets per caterpillar function. □

Proof of Theorem 10. First remove biases, if present, by taking the recession function $\lim_{t\to\infty} F(tx)/t$. This preserves the homogeneous target, the comparison graphs, and the number of neurons.

Fix an m-element label set T. Take the directional germ at $\mathbf{1}_T$, vary only the tied coordinates, and fix one tied perturbation coordinate to zero. The target germ is the support function of an $(m-1)$-simplex, whose class modulo $\mathcal{L}_{m-2}$ is nonzero. Indeed, repeated opposite-face differences and restrictions to facet coordinates would otherwise, by Equation (43), put a segment support function in $\mathcal{L}_0$; that function is not linear. The germ of the shallow remainder lies in $\mathcal{L}_1$.

For a counted unit, only comparisons with both endpoints in T remain nonlinear. Let c be the number of connected components of its induced comparison graph, including isolated vertices. The free comparison forms span a space of dimension $m-c$. If the outer preactivation is nonzero at $\mathbf{1}_T$, the germ is zero or belongs to $\mathcal{L}_1$. Otherwise, adjoining the resulting linear head gives at most $m-c+1$ directions. Signed inner weights cause no difficulty: split the local preactivation as $h_P - h_Q$, where the two polytopes lie in this span, and use $\sigma(h_P - h_Q) = h_{\mathrm{conv}(P\cup Q)} - h_Q$. Thus its germ belongs to $\mathcal{L}_{m-c+1}$. If $c \geq 3$, its class modulo $\mathcal{L}_{m-2}$ vanishes. Every tie therefore requires some counted unit whose induced graph has at most two connected components.

For a graph of maximum degree two, let C_j count its connected induced j-vertex subsets. Each path or cycle component contributes at most its number of vertices: a proper connected subset is an interval or circular arc, and a whole component contributes only once. Hence $C_j \leq N$. An m-element set with two induced components is an unordered union of two connected sets. Dropping disjointness and the absence of edges between them only increases the count. A single unit consequently covers at most

$$C_m + \frac{1}{2}\sum_{j=1}^{m-1} C_j C_{m-j} \leq N + \frac{m-1}{2}N^2 \tag{121}$$

ties of size m. Counting all $\binom{N}{m}$ ties proves Equation (36). The central binomial coefficient is $\Theta(2^N/\sqrt{N})$, giving the stated asymptotic bound. □

Proof of Theorem 11. The locality argument in the proof of Theorem 10 shows that every label set T with $|T| \geq 3$ needs a comparison graph G for which $G[T]$ has at most two components. Select each label independently with probability $1/3$. A vertex v is selected and isolated in $G[T]$ with probability $p_v = (2/3)^{\deg(v)}/3$. These events are independent for vertices in any set C with pairwise disjoint closed neighborhoods. Since $p_v/(1-p_v) \leq 1/2$, the probability of at most two such events is bounded by

$$\Pr\big(c(G[T]) \leq 2\big) \leq \left(1 + \frac{|C|}{2} + \frac{1}{4}\binom{|C|}{2}\right)\prod_{v\in C}(1-p_v) \leq a_N \prod_{v\in C}(1-p_v). \tag{122}$$

The square of a forest of maximum degree Δ has a coloring with $\Delta+1$ colors. Root each component and color in breadth-first order: only the parent, grandparent, and earlier siblings can forbid colors, at most Δ in total. Each color class has disjoint closed neighborhoods. Since a forest has at most N edges, $\prod_v p_v = 3^{-N}(2/3)^{2|E|} \geq (4/27)^N$. The arithmetic-geometric mean inequality gives first $N^{-1}\sum_v p_v \geq 4/27$, and then $\prod_v(1-p_v) \leq (1 - N^{-1}\sum_v p_v)^N \leq (23/27)^N$. The color classes partition this product, so one class C has $\prod_{v\in C}(1-p_v) \leq (23/27)^{N/(\Delta+1)}$. By Equation (122), one graph consequently serves a fraction at most $a_N(23/27)^{N/(\Delta+1)}$ of all subsets. The probability of size at least three is $1 - a_N(2/3)^N$; applying the union bound proves Equation (37).

For an arbitrary graph of maximum degree Δ, its square has maximum degree at most Δ^2 and hence admits Δ^2+1 colors. Also $p_v \geq (2/3)^{\Delta}/3$ for every vertex. A largest color class has at least $N/(\Delta^2+1)$ vertices; bound its product factor by factor and repeat the argument. Finally, repeated units with the same graph serve no additional tied sets, so both union bounds may count distinct graphs instead of units. □

Arbitrary tree shapes with a source orientation. The block argument also works with a subforest R of the reduced forest: its maximum degree must be at most two, and each vertex of Φ isolated in R must either have a pendant neighbour or belong to every passed tie set. Use the same star partition of R. The first kind of singleton has weight $1/2 + y$; the second has weight y, since it must be selected. All weights are at most one at $y = 1/6$, so the same counting argument applies. A leaf with $\kappa \notin \{0, 1\}$ is always tied: otherwise balance would require its unique edge to give an out-degree other than zero or one.

For $U_{a,s,r,T}$, root the underlying tree at the head root r. Call a nonleaf *poor* if it has neither a pendant nor an out-leaf neighbour. Choose one child of each poor vertex; such a child exists, and no child is chosen by two parents. Let R contain these chosen edges, together with every edge from an out-leaf to a nonpendant neighbour. These edges belong to the reduced forest. An isolated vertex of R in Φ is either adjacent to a pendant or a leaf with $\kappa \notin \{0, 1\}$, so it satisfies the required condition.

Leaves have degree at most one in R. A nonleaf with at most one out-leaf neighbour has degree at most two in R: a poor vertex contributes one chosen child edge and can receive one, whereas any other vertex receives at most one chosen child edge. The only remaining case is a common neighbour v of the two out-leaves s and r. It is not poor. Its parent is the leaf r, which is not poor either, so no chosen child edge ends at v. Its only R-edges are the two out-leaf edges. Thus R has maximum degree two. This gives the per-unit factor 6^m for every $U_{a,s,r,T}$, and hence the stated $(7/6)^N/(N + 1)$ bound for mixtures with the existing caterpillar and rooted-star face dictionaries.

Arbitrary orientations and large matchings. Write $n = N - 1$. For a tree G on n vertices with a matching of size k, and $y > 0$, the sum of $y^{|S|}$ over nonempty connected vertex sets S is at most $n(1 + y)^{n-2k}(1 + y + y^2)^k$. Indeed, root G at a chosen vertex v. A connected set containing v contains every ancestor of each of its vertices. On each matching edge, the state with the parent absent and the child present is forbidden. The other three states have total weight $1 + y + y^2$. Since the edges are disjoint, their weights multiply; every remaining vertex contributes $1 + y$. Summing over the n choices of v counts every nonempty connected set at least once.

Choose the common integer B in Equation (110) large enough for every unit in the enlarged finite expansion. At an m-element tie, $m \geq 3$, fix one tied perturbation coordinate to zero and consider the local functions modulo $\mathcal{L}_{m-2}$ on the remaining $m - 1$ coordinates. The maximum has the nonzero simplex class, whereas the auxiliary functions vanish. If a tree unit's outer argument is nonzero at the base point, its local function is zero or a sum of linear functions and segment support functions, so its class vanishes. If the outer argument is zero, the digit argument already used above makes every frozen coordinate coefficient zero. The apex coefficient is -1, so the apex must be tied. If the retained hinges on the other $m - 1$ tied labels are disconnected, there are at most $m - 3$ such hinges; adjoining the head gives dimension at most $m - 2$. Thus a nonzero class requires those labels to form a connected subset of the original tree. This condition is independent of edge orientation and remains necessary with omitted or linear edges.

Let K_0 count the caterpillar and rooted-star face functions, and K_1 the added functions with matching number at least k. At each permutation locality, at least one unit has a nonzero class. Such a unit satisfies the balance and activity conditions above: every nonapex tied label has a retained neighbour, and the isolated apex has coefficient -1. Consequently a function counted by K_0 contributes at most $6^m m!(N - m)!$ localities of size m. Divide the covering count by $m!(N - m)!$, multiply by y^m, and sum over $3 \leq m \leq N$. Putting $\rho = (1 + y + y^2)/(1 + y)^2$, the connected-set estimate bounds the contribution of each function counted by K_1 by $ny(1+y)^n\rho^k$; the extra factor y accounts for its fixed apex. We obtain $(1+y)^N - 1 - Ny - \binom{N}{2}y^2 \leq |K_0| \sum_{m=3}^{N} (6y)^m + |K_1| ny(1 + y)^n \rho^k$.

Take $y = 2/11$, so $\rho = 147/169$, and put $E = (13/11)^N \rho^k$ and $Z = |K_0| + |K_1|$. Since $2k \leq N - 1$ and $147 \geq 12^2$, we have $E = (13/11)^{N-2k}(147/121)^k \geq (12/11)^N \geq 1$. Also $Z \geq 1$, since the auxiliary functions alone cannot give the simplex class at a three-element tie. The preceding inequality therefore yields

$(13/11)^N \le \big(1+2N/11+4\binom{N}{2}/121+N-2+2n/13\big)EZ \le (N+1)^2EZ$. Canceling $(13/11)^N$ proves $Z \ge (169/147)^k/(N+1)^2$. Finally, the leaf criterion follows from a minimum vertex cover C of size equal to the maximum matching number ν. The nonleaf vertices outside C form an independent set I; their at least $2|I|$ incident edges lie in the forest on $I \cup C$, which has at most $|I| + \nu - 1$ edges. Hence $|I| \le \nu - 1$, and $n - L \le 2\nu - 1$.

Adding every oriented star. Work in full coordinates. For a label j, let ι_j insert a zero at that coordinate and put $\mathcal{T}_j g = (\partial_{-e_j} g - \partial_{e_j} g) \circ \iota_j$. This linear operator sends $\mathcal{L}_d$ into $\mathcal{L}_{d-1}$ for $d \ge 1$ and annihilates $\mathcal{L}_0$: if coordinate j is constant on a polytope, its two exposed faces coincide; otherwise both have smaller dimension. Projection under restriction cannot increase dimension. Moreover, $\mathcal{T}_j \max_m = \max_{m-1}$, since the simplex faces are its facet opposite e_j and the vertex e_j.

For a canonical star, deletion of a nonroot leaf gives the smaller star modulo $\mathcal{L}_1$, with sign $+$ for $\sigma(x_l - x_c)$ and sign $-$ for $\sigma(x_c - x_l)$. If the deleted leaf is the head root, the latter orientation transfers the head to the centre with sign $+$; the former gives zero modulo $\mathcal{L}_1$. Deleting the apex leaves a zonotope, and deleting the centre fixes every segment, so these cases also give zero modulo $\mathcal{L}_1$. Thus successive deletions preserve the span of canonical stars modulo the lower-dimensional terms.

At five labels, a six-flag certificate gives a linear functional that annihilates every canonical oriented star and $\mathcal{L}_3$, and takes value 120 on the simplex. The independent vertex checker checks all 640 labelled stars, with every head root and orientation, by successive maxima and minima over explicit vertex sets. Composing this functional with $m - 5$ coordinate deletions therefore gives such a separator at every $m \ge 5$. A nonhinge state already has dimension at most $m - 2$, so all star face states are covered too.

Assign every term whose function belongs to the star family to the added star class, even if it was originally listed in another class. Here K_0, K_1 count only the remaining terms, and $Z = |K_0| + |K_1|$. At any tie of size at least five, a surviving top-dimensional star germ is again a canonical star: each untied component has a unique attachment to the tied subtree, and summing its zero frozen coefficients transports the head to that attachment if it contains the original root. Apply the separator to the local expansion. Every added star term vanishes, so a term from the other classes must cover the locality. The preceding weighted count now starts at $m = 5$. With caterpillars alone, $y = 1/6$ gives $(7/6)^N - \sum_{j=0}^{4}\binom{N}{j}6^{-j} \le (N-4)|K_0|$. There is at least one nonstar term, and $\sum_{j=0}^{4}\binom{N}{j}6^{-j} + N - 4 \le (N+1)^4$ for $N \ge 5$, which proves the first assertion. For the matching class, the same choice $y = 2/11$ and the same E as above give $(13/11)^N \le \big(\sum_{j=0}^{4}\binom{N}{j}(2/11)^j + N - 4 + 2n/13\big)EZ \le (N+1)^4EZ$. This proves the stated bound without counting the added star terms.

C.2 Proof of the dimension bound

Proof of Theorem 12. Put $r = N - 2$ and $b = N - 1$. We exhibit $r!$ dictionary functions and linear functionals whose evaluation matrix is the identity and which annihilate A_N.

Step 12.1 (One path for each order). For every permutation π fixing 0 and b, use the path $\pi(1), \dots, \pi(r), b$ with apex 0 and keep every edge as a hinge:

$$U_\pi(x) = \sigma\Big(x_b - x_0 + \sum_{j=1}^{r} \sigma(x_{\pi(j)} - x_{\pi(j+1)})\Big) \in V_N. \tag{123}$$

There are $r!$ such paths.

Step 12.2 (One mixed difference). For an order ρ, set $c = \rho(r)$ and

$$z_0^\rho = z_b^\rho = 2r, \qquad z_{\rho(j)}^\rho = 2j \quad (1 \le j \le r). \tag{124}$$

Thus $0, b, c$ tie at the top. Let $\Lambda_\rho g = \sum_{S\subseteq\{0,b,c\}} (-1)^{3-|S|} g(z^\rho + \mathbf{1}_S)$. This third mixed difference annihilates every function of at most two coordinates, by pairing corners along a missing coordinate. In particular, $\Lambda_\rho A_N = 0$.

Step 12.3 (The evaluation matrix is the identity). On the cube of Λ_ρ, every inner function of U_ρ except the last is zero: adjacent base heights differ by 2, while a coordinate moves by at most 1. Writing $v = x - z^\rho$, we obtain $U_\rho = \sigma(v_b - v_0 + \sigma(v_c - v_b))$. At the eight corners this is 1 exactly when $v_0 = 0$ and $(v_b, v_c) \neq (0, 0)$, so its mixed difference is $1 + 1 - 1 = 1$.

If $\pi \neq \rho$, the middle labels in the π-order have a descent in ρ-height. The corresponding inner function is at least 1 everywhere on the cube, whereas $x_b - x_0 \geq -1$. The outer ReLU can therefore be removed there. The resulting sum depends on at most two coordinates term by term, so its third mixed difference is zero. Hence

$$\Lambda_\rho U_\pi = \begin{cases} 1, & \pi = \rho, \\ 0, & \pi \neq \rho. \end{cases} \tag{125}$$

Step 12.4 (Independence modulo the auxiliary span). Applying Λ_ρ to $\sum_\pi c_\pi U_\pi \in A_N$ gives $c_\rho = 0$. Thus these $r!$ functions are independent modulo A_N, proving $q_N \geq (N-2)!$. □

D Relation to the construction of Brunck et al.

The combinatorial proof of [BHR+26, Part 2] cones zonotopes $B_{v,\pi,M}$ to pyramids and expresses the simplex support function as a signed sum of their support functions modulo lower-dimensional terms [BHR+26, Lemma 22]. This appendix identifies these building blocks. For a nonempty set M of marked positions, the pyramid has the support function of a translated caterpillar polytope, with apex v, label order π, and cut set $M - 1$. For $M = \varnothing$ it gives a rooted star, which is not a caterpillar function. We use the indexing of [BHR+26, Section 7.1], with their n written as N and $[N]$ identified with our label set $\{0, \ldots, N-1\}$. Positions of an ordering are numbered from 1.

Definition 17 (Zonotopes of Brunck et al., `Lean_31`, `Lean_32`)**.** Let $N \geq 3$, let v be a label, let $\pi = (\pi_1, \ldots, \pi_{N-1})$ be an ordering of the remaining labels, and let $M \subseteq \{2, \ldots, N-1\}$. For $p \in \mathbb{R}^N$ put $u_i(p) := \sum_{1\leq j<i} p_{\pi_j}$. Following [BHR+26, Equation (3)], define

$$H_{-v} := \Big\{ p \in \mathbb{R}^N : p_v = 0,\ \sum_k p_k = 1 \Big\}, \tag{126}$$

$$B_{v,\pi,M} := \left\{ p \in H_{-v} : \begin{array}{ll} 0 \leq u_i(p) \leq 1 & \text{for } i \in M, \\ -1 \leq p_{\pi_i} \leq 0 & \text{for } i \in \{2, \ldots, N-1\} \setminus M \end{array} \right\}. \tag{127}$$

The pyramid with apex e_v is $\mathrm{pyr}_{e_v}(B_{v,\pi,M}) := \mathrm{conv}(\{e_v\} \cup B_{v,\pi,M})$, with support function $h_{v,\pi,M}$ as in Equation (16). Its last marked position and the marked parent of a position i are

$$q := \max(\{1\} \cup M), \qquad r(i) := \max\big(\{1\} \cup \{j \in M : j < i\}\big) \qquad (2 \leq i \leq N-1). \tag{128}$$

For $M \neq \varnothing$, $q = \max M$. For labels $a \neq c$, the *rooted star* with apex a and root c is

$$S_{a,c}(x) := \sigma\Big(x_c - x_a + \sum_{l \notin \{a,c\}} \sigma(x_c - x_l)\Big). \tag{129}$$

Theorem 18 (Relation to Brunck et al., `Lean_33`, `Lean_34`, `Lean_35`, `Lean_36`, `Lean_37`, `Lean_38`, `Lean_39`, `Lean_40`, Proof)**.** *Let v, π, and M be as in Definition 17, and let q and $r(i)$ be as in Equation* (128)*.*

(a) The base is the zonotope

$$B_{v,\pi,M} = e_{\pi_q} + \sum_{i=2}^{N-1} \big[0, e_{\pi_{r(i)}} - e_{\pi_i}\big]. \tag{130}$$

(b) For every $x \in \mathbb{R}^N$,

$$h_{v,\pi,M}(x) = x_v + \sigma\Big(x_{\pi_q} - x_v + \sum_{i=2}^{N-1} \sigma\big(x_{\pi_{r(i)}} - x_{\pi_i}\big)\Big). \tag{131}$$

(c) Let $M \neq \varnothing$. Define the caterpillar description (π', S) by $\pi'(0) := v$, $\pi'(j) := \pi_j$ for $1 \le j \le N-1$, and $S := \{i-1 : i \in M\}$. Then $|S| = |M|$ and

$$h_{v,\pi,M}(x) = x_v + U_{\pi',S}(x) \qquad (x \in \mathbb{R}^N). \tag{132}$$

Hence $h_{v,\pi,M} - x_v = U_k$ for some $k \in \mathcal{K}_0$. Conversely, every caterpillar description (π', S) arises in this way from $v = \pi'(0)$, $\pi_j = \pi'(j)$, and $M = \{j+1 : j \in S\} \neq \varnothing$.

(d) Let $M = \varnothing$. Then $h_{v,\pi,\varnothing}(x) = x_v + S_{v,\pi_1}(x)$. No rooted star is a caterpillar function: $S_{a,c} \neq U_k$ for all labels $a \neq c$ and all $k \in \mathcal{K}_0$. In particular, $h_{v,\pi,\varnothing} - x_v \neq U_k$ for every $k \in \mathcal{K}_0$.

By Equation (17), the right-hand side of Equation (132) is the support function of the translated caterpillar polytope $e_v + Q_{h_{\pi',S},L_{\pi',S}}$. The family of [BHR+26] therefore consists of the caterpillar polytopes, translated by e_v, and of the pyramids over rooted stars. The sign $(-1)^{|M|}$ of [BHR+26, Lemma 22] equals $(-1)^{|S|}$. The theorem compares building blocks only. It makes no statement about corrections or network coefficients.

Proof of Theorem 18. Write $I := \{2, \dots, N-1\}$ and $g_i := e_{\pi_{r(i)}} - e_{\pi_i}$ for $i \in I$. Every element of $\{1\} \cup M$ lies in $\{1, \dots, N-1\}$, so q and all $r(i)$ are positions. Since $M \subseteq I$, the definitions give

- $1 \le r(i) < i$ for $i \in I$, and $i \le r(l)$ whenever $i \in M$ and $i < l$,
- $i \le q$ for $i \in M$,
- $q \neq i$ and $r(l) \neq i$ for $i \in I \setminus M$ and all l.

Step 18.1 (Coordinates of the zonotope). For $t \in [0,1]^I$ let $z(t) := e_{\pi_q} + \sum_{l \in I} t_l g_l$. Every e_{π_j} has coordinate sum 1 and vanishes at v, so $z(t) \in H_{-v}$. For $i \in I \setminus M$ the coordinate $z(t)_{\pi_i}$ collects the term $-t_i$ and no other term, since π_i is neither π_q nor any $\pi_{r(l)}$. Hence $z(t)_{\pi_i} = -t_i$. For $i \in M$, the prefix sum u_i counts a vector e_{π_j} exactly when $j < i$. Thus e_{π_q} contributes 0, because $q \ge i$. A generator g_l with $l < i$ contributes $1 - 1 = 0$, because $r(l) < l < i$. A generator with $l > i$ contributes $0 - 0$, because $r(l) \ge i$. Finally g_i contributes t_i, because $r(i) < i$. Hence $u_i(z(t)) = t_i$. Therefore $z(t) \in B_{v,\pi,M}$.

Step 18.2 (The coordinates determine the point). Let $p, p' \in H_{-v}$ with $u_i(p) = u_i(p')$ for $i \in M$ and $p_{\pi_i} = p'_{\pi_i}$ for $i \in I \setminus M$. Put $d := p - p'$ and $U_i := \sum_{1 \le j < i} d_{\pi_j}$. Then $U_1 = 0$, and $U_N = \sum_k d_k - d_v = 0$. Descending from $i = N-1$ to $i = 2$, either $i \in M$ and $U_i = 0$ by assumption, or $i \notin M$ and $U_i = U_{i+1} - d_{\pi_i} = U_{i+1} = 0$. Hence $d_{\pi_j} = U_{j+1} - U_j = 0$ for $1 \le j \le N-1$, and $d_v = 0$. Thus $p = p'$. This is the invertibility in [BHR+26, Claim 21].

Step 18.3 (Proof of (a)). Step 18.1 shows that the right-hand side of Equation (130) lies in $B_{v,\pi,M}$. Conversely, let $p \in B_{v,\pi,M}$, and set $t_i := u_i(p)$ for $i \in M$ and $t_i := -p_{\pi_i}$ for $i \in I \setminus M$. The inequalities in Equation (127) say $t \in [0,1]^I$. By Step 18.1, the points p and $z(t)$ of H_{-v} have the same coordinates, so $p = z(t)$ by Step 18.2.

Step 18.4 (Proof of (b)). A linear function attains the same supremum on a set and on its convex hull. Hence $h_{v,\pi,M}(x) = \max(x_v, \sup_{p \in B_{v,\pi,M}} \langle p, x \rangle)$. Put $b_l := x_{\pi_{r(l)}} - x_{\pi_l}$. By (a), $\langle z(t), x \rangle = x_{\pi_q} + \sum_l t_l b_l$, and $t_l b_l \le \sigma(b_l)$ for $0 \le t_l \le 1$, with equality for $t_l = \mathbb{1}_{\{b_l \ge 0\}}$. So the supremum over $B_{v,\pi,M}$ equals $A := x_{\pi_q} + \sum_l \sigma(b_l)$. Then Equation (7) gives $\max(x_v, A) = x_v + \sigma(A - x_v)$, which is Equation (131).

Step 18.5 (Proof of (c)). With π' and S as in (c), position j of the caterpillar carries the label $\pi'(j+1) = \pi_{j+1}$, so caterpillar position j is the position $j+1$ of [BHR+26]. The map $i \mapsto i - 1$ is a bijection from M onto S, hence $|S| = |M|$, and it carries $\{1\} \cup M$ onto $\{0\} \cup S$ and $\{1\} \cup \{j \in M : j < i\}$ onto $\{0\} \cup \{j \in S : j < i - 1\}$. Comparing with Equation (10), the last spine position is $q - 1$ and the parent of caterpillar position $i - 1$ is $r(i) - 1$. By Equations (12) and (13), the head is $x_{\pi_q} - x_v$ and the edge functions are $x_{\pi_{r(i)}} - x_{\pi_i}$, $i \in I$. So the right-hand side of Equation (131) is $x_v + U_{\pi',S}(x)$. If $M \neq \varnothing$, then $S \neq \varnothing$ and $0 \notin S$, so (π', S) is a caterpillar description, and its normalized description $k \in \mathcal{K}_0$ satisfies $U_k = U_{\pi',S}$ by Equation (8). The converse inverts these bijections.

Step 18.6 (Proof of (d)). For $M = \varnothing$, $q = 1$ and $r \equiv 1$, and $\{\pi_i : i \in I\}$ is the set of labels other than v and π_1. So Equation (131) becomes $x_v + S_{v,\pi_1}(x)$. Let $a \neq c$ and evaluate at $x = e_c$. Every inner term of $S_{a,c}(e_c)$ equals $\sigma(1)$, so $S_{a,c}(e_c) = 1 + (N-2) = N - 1$. Let $k \in \mathcal{K}_0$. Then U_k is the normalization of a face function $U^{\tau}_{\pi',S}$, and $U_k = U^{\tau}_{\pi',S}$ by Equation (8). Since $\ell \le \sigma \circ \ell$, $0 \le \sigma \circ \ell$, and σ is monotone, $U^{\tau}_{\pi',S} \le U_{\pi',S}$. At $x = e_c$, every head and edge function takes values in $\{-1, 0, 1\}$. If $c = \pi'(0)$, the argument of the outer σ in $U_{\pi',S}(e_c)$ is -1, so $U_{\pi',S}(e_c) = 0$. Otherwise c is the label of a caterpillar position m. The head contributes at most 1, and every inner hinge at most 1. If $m \ge 1$, the inner hinge of the edge into position m is $\sigma(x_{\pi'(p(m)+1)} - 1) = 0$, so the value is at most $1 + (N-3)$. If $m = 0$, the head contributes 0, because $q \ge \min S > 0$, so the value is at most $N - 2$. Thus $U_k(e_c) \le N - 2 < S_{a,c}(e_c)$. □

E Optimal second width for the prescribed five-input layer

Proof of Theorem 16. Let $u = x_0 - x_1$, $v = x_2 - x_1$, $w = x_3 - x_1$, and $z = x_4 - x_1$. Write $A = \max\{u, v\}$, $B = \max\{w, z\}$, $S = a + b$, $T = c + d$, $D = |a - b|$, and $E = |c - d|$. Thus $\eta = B - A$. The elementary identity $\sigma(p+q) + \sigma(p-q) = p + \max\{|p|, |q|\}$ shows that the right side of Equation (78), minus x_1, is

$$\frac{S + \max\{D, E\}}{2} + \sigma(B - A) - \sigma\left(B - A + \frac{S - T}{2}\right). \tag{133}$$

If $A, B \ge 0$, then $2A = S + D$ and $2B = T + E$, so Equation (133) reduces to $A + \sigma(B - A) = \max\{A, B\}$. If $A \le 0 \le B$, then $S = D = 0$ and $B - A \ge B \ge T/2$; the difference of the two ReLUs is $T/2$, giving B. If $B \le 0 \le A$, both ReLUs vanish and the expression is A. If $A, B \le 0$, then $S = T = D = E = 0$ and the expression is zero. This proves Equation (78) on all real inputs and establishes the upper bound.

For the lower bound, let W be the span of the first eight functions in Equation (77), and put $f = \max_5 - x_1$. Let k_W be the least K for which

$$f = R + \sum_{j=1}^{K} d_j \sigma(g_j), \qquad R, g_j \in W. \tag{134}$$

The minimum second hidden layer width for the prescribed first hidden layer is $k_W + 2$. To see this, every second-layer preactivation has the form $v_j + \alpha_j \sigma(x_1) + \beta_j \sigma(-x_1)$ with $v_j \in W$. Translate every input

coordinate by the same scalar. At positive infinity, the nonlinear terms that remain have $\alpha_j = 0$, so there must be at least k_W such neurons, as well as a passing neuron accounting for the linear growth. The negative end gives the same conclusion for β_j.

If there were only $k_W + 1$ neurons, each end would have a unique passing neuron. Call them n, m; their output coefficients have signs $d_n > 0 > d_m$, so they are distinct. All other neurons have both slopes zero, and $\beta_n = \alpha_m = 0$. The two translation identities are therefore

$$f = d_n v_n + d_m \sigma(v_m) + C = d_n \sigma(v_n) + d_m v_m + C, \tag{135}$$

where C consists of the $k_W - 1$ common nonlinear terms. Subtracting gives $d_n \sigma(-v_n) = d_m \sigma(-v_m)$, so both sides vanish. Absorbing the two passing terms into W contradicts the minimality of k_W. Conversely, any representation in Equation (134) is compiled with $K + 2$ neurons by representing $x_1 + R$ as a difference of two ReLUs. No assumption that W contains all linear functions is used.

It remains to establish $k_W \geq 4$. The following exact finite certificate excludes $K \leq 3$; its domain includes all real coefficients. Absorb each nonzero output magnitude into its preactivation, so $d_j \in \{-1, 1\}$. For a coordinate subset U, write

$$Q_U(g) = \sum_{\varnothing \neq S \subsetneq U} (-1)^{|U|-|S|} \, |g(\mathbf{1}_S)|. \tag{136}$$

On Boolean indicator vectors, every $g \in W$ has degree at most two and vanishes on the empty and full vectors. At a generic point of a top triple tie U with next label o, only terms satisfying $g_j(\mathbf{1}_U) = g_j(\mathbf{1}_{U \cup \{o\}}) = 0$ contribute to its third mixed difference. At a top four-label tie U, the condition is $g_j(\mathbf{1}_U) = 0$; at the full tie every term contributes. For each of these localities, taking the mixed difference in Equation (134) and using $2\sigma(g) = g + |g|$ gives

$$\sum_{j \text{ satisfying the stated vanishing conditions}} d_j Q_U(g_j) = 2(-1)^{|U|+1}. \tag{137}$$

There are twenty triple equations, five four-label equations, and one full-tie equation. The remainder is annihilated by these differences.

Seven further equations retain the restriction $R \in W$. Evaluate the eight first hidden layer functions spanning W on the fifteen singleton and pair Boolean rays. The resulting 15×8 matrix has rank eight. For every vector λ in its left kernel, Equation (134) implies

$$\sum_{1 \leq |S| \leq 2} \lambda_S \sum_j d_j |g_j(\mathbf{1}_S)| = 2 \sum_{1 \leq |S| \leq 2} \lambda_S \big(1 - \mathbf{1}_{1 \in S}\big). \tag{138}$$

Here $[5] = \{0, 1, 2, 3, 4\}$.

The fifteen linear forms $g \mapsto g(\mathbf{1}_U)$ with $|U| = 3, 4$ have 623 exact vanishing flats in W, including the zero flat. Every nonzero real preactivation lies in one of the other 622 strata. The checker regenerates these flats by intersections of kernels and checks which equations in Equation (137) can receive a nonzero contribution. There are 30,041 unordered covering triples of flats and eight output-sign choices for each, giving 240,328 cases. The parameter space in each case is the entire corresponding linear flat: no coefficient grid or magnitude bound is imposed. Cases with extra vanishing forms are also included under their own exact strata. Repeated flats are included. A representation with one or two nonzero nonlinear terms can be padded to three by splitting a term into two equal positive multiples. With no nonlinear terms, the full-tie equation would give $0 = 2$. Thus these cases cover every $K \leq 3$.

For 239,996 cases the certificate gives a rational separation using Equations (137) to (138), nonnegativity of absolute values, and triangle inequalities from explicitly verified linear dependencies among their arguments.

The other 332 cases have finite trees that branch on both closed signs of an argument and end in rational linear contradictions. The independent checker rebuilds the unreduced equations, checks every dependency and rational contradiction, and verifies that all cases and all branches are covered. The known four-term identity satisfies the same equations as a positive control. Thus $k_W \geq 4$, while Equation (78) gives $k_W \leq 4$. The required width is $k_W + 2 = 6$.

Finally, biases cannot lower this bound. For any exact biased network, the recession limit $\lim_{t\to\infty} F(tx)/t$ removes all biases and preserves its input weight rows and both widths; the target is unchanged by positive homogeneity. □

The certificate description, domain checker, rational certificate checker, and verification receipt are pinned to the same repository snapshot. The compressed certificate is 1.5 MiB. These checks use exact rational arithmetic; they are separate from the Lean verification discussed in Appendix F.

F Lean definitions and main theorem

The real parameter arrays in Equation (4) have dimensions

$$\begin{aligned} W_1 &\in \mathbb{R}^{h_1 \times N}, \quad b_1 \in \mathbb{R}^{h_1}, \\ W_2 &\in \mathbb{R}^{h_2 \times h_1}, \quad b_2 \in \mathbb{R}^{h_2}, \\ w_3 &\in \mathbb{R}^{h_2}, \qquad b_3 \in \mathbb{R}, \end{aligned} \tag{139}$$

Sources (`Definitions.lean`, `Theorems.lean`). The listings specify the network, widths, and correctness for every real input. They reproduce declarations from these files at revision `8b4d74ab66f4`, omitting imports, documentation comments, and the theorem's proof body; these remain in the linked sources.

F.1 Network and width definitions

```
namespace MaxCaterpillar

open BigOperators

abbrev Point (N : ℕ) := Fin N → ℝ

noncomputable def relu (t : ℝ) : ℝ := max 0 t

structure Network (N h₁ h₂ : ℕ) where
  W₁ : Fin h₁ → Fin N → ℝ
  b₁ : Fin h₁ → ℝ
  W₂ : Fin h₂ → Fin h₁ → ℝ
  b₂ : Fin h₂ → ℝ
  w₃ : Fin h₂ → ℝ
  b₃ : ℝ

noncomputable def Network.eval {N h₁ h₂ : ℕ} (net : Network N h₁ h₂)
    (x : Point N) : ℝ :=
  (∑ k, net.w₃ k * relu
    ((∑ j, net.W₂ k j * relu
```

```
      ((∑ i, net.W₁ j i * x i) + net.b₁ j)) + net.b₂ k))
    + net.b₃

def Network.HasRationalWeights {N h₁ h₂ : ℕ}
    (net : Network N h₁ h₂) : Prop :=
  (∀ j i, ∃ q : ℚ, (q : ℝ) = net.W₁ j i) ∧
    (∀ k j, ∃ q : ℚ, (q : ℝ) = net.W₂ k j) ∧
    ∀ k, ∃ q : ℚ, (q : ℝ) = net.w₃ k

def Network.HasZeroBiases {N h₁ h₂ : ℕ}
    (net : Network N h₁ h₂) : Prop :=
  net.b₁ = 0 ∧ net.b₂ = 0 ∧ net.b₃ = 0

noncomputable def Maximum (N : ℕ) [NeZero N] (x : Point N) : ℝ :=
  Finset.univ.sup'
    ⟨⟨0, Nat.pos_of_ne_zero (NeZero.ne N)⟩,
      Finset.mem_univ _⟩ x

def firstWidth (N : ℕ) : ℕ := N.choose 2 + N + 1

def coneCount (N : ℕ) : ℕ :=
  ∑ m in Finset.Icc 2 (N - 1),
    N.choose m * (m.factorial / 2) * (N - m)
      * m ^ (N - m - 1)

def secondWidth (N : ℕ) : ℕ :=
  coneCount N * 3 ^ (N - 2) - N * (N - 1) + 2

end MaxCaterpillar
```

F.2 Caterpillar functions and expansions

These definitions, also in `Definitions.lean`, state the results of Section 4. A description $k = (h, M)$ is stored in reduced coordinates, and `realUnitY` evaluates U_k on $\mathbb{R}^N$.

```
abbrev FaceKey (r : ℕ) :=
  MaxNN.Caterpillar.Corrections.RealKey (r + 1)

noncomputable def caterpillarDescriptions (r : ℕ) :
    Finset (FaceKey r) :=
  MaxNN.Caterpillar.Corrections.normalizedKeys r

noncomputable def caterpillarFunction {r : ℕ}
    (k : FaceKey r) : Point (r + 1 + 1) → ℝ :=
  MaxNN.Caterpillar.Corrections.realUnitY k

noncomputable def auxiliaryFunction (N : ℕ)
    (j : Fin (MaxNN.Width.W N)) : Point N → ℝ :=
```

```
  fun x => MaxNN.Width.feat N x j

def IsCaterpillarExpansion (r : ℕ)
    (K : Finset (FaceKey r)) : Prop :=
  K ⊆ caterpillarDescriptions r ∧
    ∃ (d : FaceKey r → ℝ)
      (a : Fin (MaxNN.Width.W (r + 1 + 1)) → ℝ),
      ∀ x, Maximum (r + 1 + 1) x =
        ∑ k in K, d k * caterpillarFunction k x +
          ∑ j, a j * auxiliaryFunction (r + 1 + 1) j x

noncomputable def auxiliarySpan (N : ℕ) :
    Submodule ℝ (Point N → ℝ) :=
  Submodule.span ℝ (Set.range (auxiliaryFunction N))

noncomputable def caterpillarSpan (r : ℕ) :
    Submodule ℝ (Point (r + 1 + 1) → ℝ) :=
  Submodule.span ℝ
    ((caterpillarFunction '' ↑(caterpillarDescriptions r))
      ∪ Set.range (auxiliaryFunction (r + 1 + 1)))

noncomputable def caterpillarDim (r : ℕ) : ℕ :=
  FiniteDimensional.finrank ℝ (↑(caterpillarSpan r) /
    (auxiliarySpan (r + 1 + 1)).comap
      (caterpillarSpan r).subtype)
```

F.3 The main theorems

The assumption `[NeZero N]` expresses positive arity. For $N \geq 3$ the first statement is Theorem 1; the natural-number width definitions also cover $N = 1, 2$. With $N = r + 2$, the next three are Theorem 8, and the last is Theorem 12. All are in `Theorems.lean`.

```
theorem caterpillar_net (N : ℕ) [NeZero N] :
    ∃ net : Network N (firstWidth N) (secondWidth N),
      (∀ x, net.eval x = Maximum N x) ∧
        net.HasRationalWeights ∧ net.HasZeroBiases

theorem caterpillar_expansion_choose_le (r : ℕ)
    {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K)
    {k : ℕ} (hk : 3 ≤ k) (hkN : k ≤ r + 1 + 1) :
    (r + 1 + 1).choose k ≤ 6 ^ k * K.card

theorem caterpillar_expansion_exponential (r : ℕ)
    (hr : 1 ≤ r) {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K) :
    7 ^ (r + 1 + 1) ≤
      (r + 1 + 1 + 1) * 6 ^ (r + 1 + 1) * K.card
```

```
theorem caterpillar_expansion_quadratic (r : ℕ)
    (hr : 1 ≤ r) {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K) :
    (r + 1 + 1) * (r + 1) ≤ 6 * K.card

theorem caterpillarDim_ge (r : ℕ) (hr : 1 ≤ r) :
    r.factorial ≤ caterpillarDim r
```

F.4 The building blocks of Brunck et al.

The definitions of Definition 17, in the indexing of [BHR+26, Section 7.1], are in `Competing/Defs.lean`. An ordering is a bijection from the $N-1$ positions onto the labels other than v, and `Ordering.pos` returns π_i at position i, $1 \le i \le N-1$.

```
noncomputable def supportFunction {n : ℕ}
    (P : Set (Fin n → ℝ)) (y : Fin n → ℝ) : ℝ :=
  sSup ((fun p => ∑ k, p k * y k) '' P)

def pyramid {n : ℕ} (q : Fin n → ℝ)
    (P : Set (Fin n → ℝ)) : Set (Fin n → ℝ) :=
  convexHull ℝ (insert q P)

def zonotope {n : ℕ} (a : Fin n → ℝ) (I : Finset ℕ)
    (g : ℕ → Fin n → ℝ) : Set (Fin n → ℝ) :=
  {p | ∃ t : ℕ → ℝ, (∀ i ∈ I, 0 ≤ t i ∧ t i ≤ 1) ∧
    p = a + ∑ i in I, t i • g i}

abbrev BHRSY.Ordering (n : ℕ) (v : Fin n) :=
  Fin (n - 1) ≃ {k : Fin n // k ≠ v}

def BHRSY.Ordering.pos {v : Fin n} (π : Ordering n v)
    (i : ℕ) : Fin n :=
  if h : i - 1 < n - 1 then (π ⟨i - 1, h⟩).1 else v

def BHRSY.hyperplane (v : Fin n) : Set (Fin n → ℝ) :=
  {p | p v = 0 ∧ ∑ k, p k = 1}

noncomputable def BHRSY.prefixSum {v : Fin n}
    (π : Ordering n v) (p : Fin n → ℝ) (i : ℕ) : ℝ :=
  ∑ j in Finset.Ico 1 i, p (π.pos j)

def BHRSY.parallelepiped (v : Fin n) (π : Ordering n v)
    (M : Finset ℕ) : Set (Fin n → ℝ) :=
  {p | p ∈ hyperplane v ∧
    (∀ i ∈ M, 0 ≤ prefixSum π p i ∧ prefixSum π p i ≤ 1) ∧
    ∀ i ∈ Finset.Icc 2 (n - 1), i ∉ M →
      -1 ≤ p (π.pos i) ∧ p (π.pos i) ≤ 0}
```

```
def BHRSY.paramid (v : Fin n) (π : Ordering n v)
    (M : Finset ℕ) : Set (Fin n → ℝ) :=
  pyramid (Pi.single v 1) (parallelepiped v π M)

noncomputable def BHRSY.paramidSupport (v : Fin n)
    (π : Ordering n v) (M : Finset ℕ) : (Fin n → ℝ) → ℝ :=
  supportFunction (paramid v π M)

def BHRSY.lastMarked (M : Finset ℕ) : ℕ :=
  (insert 1 M).max' (Finset.insert_nonempty _ _)

def BHRSY.markedParent (M : Finset ℕ) (i : ℕ) : ℕ :=
  (insert 1 (M.filter fun j => j < i)).max'
    (Finset.insert_nonempty _ _)
```

The dictionary uses the word $\pi' = (v, \pi_1, \ldots, \pi_{N-1})$ (`Ordering.word`), the cut set $S = M - 1$ (`cutSet`), and the description $(h_{\pi',S}, L_{\pi',S})$ of a word (`caterpillarWordDescription`). The rooted star is `BHRSY.rootedStar`. With $N = r + 2$, `Theorems.lean` states Theorem 18 as follows.

```
theorem bhrsy_parallelepiped_eq_zonotope {r : ℕ}
    {v : Fin (r + 1 + 1)} (π : BHRSY.Ordering (r + 1 + 1) v)
    {M : Finset ℕ} (hM : M ⊆ Finset.Icc 2 (r + 1)) :
    BHRSY.parallelepiped v π M =
      zonotope (Pi.single (π.pos (BHRSY.lastMarked M)) 1)
        (Finset.Icc 2 (r + 1))
        (fun i => Pi.single (π.pos (BHRSY.markedParent M i)) 1
          - Pi.single (π.pos i) 1)

theorem bhrsy_support_eq {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) {M : Finset ℕ}
    (hM : M ⊆ Finset.Icc 2 (r + 1)) (y : Point (r + 1 + 1)) :
    BHRSY.paramidSupport v π M y =
      y v + MaxCaterpillar.relu
        (y (π.pos (BHRSY.lastMarked M)) - y v +
          ∑ i in Finset.Icc 2 (r + 1), MaxCaterpillar.relu
            (y (π.pos (BHRSY.markedParent M i)) - y (π.pos i)))

theorem bhrsy_support_eq_word {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) {M : Finset ℕ}
    (hM : M ⊆ Finset.Icc 2 (r + 1)) :
    (BHRSY.cutSet (r := r) M).card = M.card ∧
      ∀ y : Point (r + 1 + 1), BHRSY.paramidSupport v π M y =
        y v + caterpillarFunction
          (caterpillarWordDescription π.word (BHRSY.cutSet M)) y

theorem bhrsy_support_mem_caterpillar {r : ℕ}
    {v : Fin (r + 1 + 1)} (π : BHRSY.Ordering (r + 1 + 1) v)
    {M : Finset ℕ} (hM : M ⊆ Finset.Icc 2 (r + 1))
```

```
    (hne : M.Nonempty) :
    ∃ k ∈ caterpillarDescriptions r, ∀ y,
      BHRSY.paramidSupport v π M y = y v + caterpillarFunction k y

theorem bhrsy_support_empty {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) (y : Point (r + 1 + 1)) :
    BHRSY.paramidSupport v π ∅ y =
      y v + BHRSY.rootedStar v (π.pos 1) y

theorem rootedStar_ne_caterpillarFunction {r : ℕ}
    {a c : Fin (r + 1 + 1)} (hac : a ≠ c) {k : FaceKey r}
    (hk : k ∈ caterpillarDescriptions r) :
    BHRSY.rootedStar a c ≠ caterpillarFunction k
```

The same file contains the converse `bhrsy_word_mem`, the special cases `bhrsy_parallelepiped_eq_zonotope_of_nonempty` and `bhrsy_parallelepiped_empty`, the statement `bhrsy_support_empty_not_caterpillar`, and the dichotomy `bhrsy_family`.

F.5 Proof routes in the formalization

The public theorem `caterpillar_net` uses the rational projection argument directly. The essential normalized descriptions and auxiliary functions span the maximum; applying a $\mathbb{Q}$-linear projection gives rational coefficients on the same support, and the compilation theorem produces the network with rational weights and zero biases. The separate compression theorem first chooses a basis modulo the auxiliary span, giving the second width $q_N + 2$.

The count argument factors over stars of a reduced forest, and the vocabulary-dimension argument separates path functions by a third mixed difference. These are the arguments presented in Appendices C.1 to C.2.

The keyword `noncomputable` allows classical definitions of real quantities. The theorem is an exact representation statement; it does not by itself assert an efficient algorithm for extracting numerical weights.

F.6 Verification scope

The repository's historical verification certificate, dated 24 October 2026, records a build with Lean 4.4.0 and checks 122 public statements against their exact types. Its permitted axioms are `propext`, `Classical.choice`, and `Quot.sound`. The checked interface includes dictionary sufficiency, rational expansions, network correctness with rational weights and zero biases, the width bounds, the compilation and compression theorems, the small-arity networks, the Wang–Sun and Koutschan–Moser–Ponomarchuk–Schicho representations, and the network theorems for sums of maxima and CPWL functions with their rationality and bias statements. The equation and theorem links identify supporting declarations for the mathematical arguments in the paper. The certificate and its source hashes are available in the linked verification record. The subsequent consolidation of the Lean modules is not covered by that certificate: the current sources require an updated exact-statement manifest and a fresh verification run. In particular, the source-orientation, matching-number, and arbitrary-star extensions are mathematical arguments in this paper, not claims checked by the historical certificate. The mathematical arguments are provided in the main text and Appendices A to C; the formalization is an additional certificate of the formal statements.

AI methodology statement

Claude Code and Codex were used for mathematical exploration and proof search, development of the Lean formalization, and drafting and revision of the manuscript. Their use materially affected the construction and proofs in Sections 3 to 8 and Appendices A to F. The author set the research objectives, directed the searches, and revised the mathematical presentation. The accompanying Lean certificate checks the formal statements and their axiom dependencies; it does not certify the prose exposition, literature comparison, or claims of originality. The author is responsible for the contents of the paper.

References

[ABMM18] Raman Arora, Amitabh Basu, Poorya Mianjy, and Anirbit Mukherjee. Understanding deep neural networks with rectified linear units. In *International Conference on Learning Representations*, 2018. arXiv:1611.01491v6.

[AHM25] Gennadiy Averkov, Christopher Hojny, and Maximilian Merkert. On the expressiveness of rational ReLU neural networks with bounded depth. In *International Conference on Learning Representations*, 2025. OpenReview.

[BBH+26] Egor Bakaev, Florestan Brunck, Christoph Hertrich, Jack Stade, and Amir Yehudayoff. Better neural network expressivity: Subdividing the simplex. In *Proceedings of the 58th Annual ACM Symposium on Theory of Computing*, pages 500–507, 2026. doi:10.1145/3798129.3800768.

[BHR+26] Florestan Brunck, Christoph Hertrich, Kilian Rueß, Jack Stade, and Amir Yehudayoff. Two ReLU layers is all you need. Preprint, arXiv:2610.28587, version 1, October 2026. arXiv:2610.28587v1.

[Fre42] Hans Freudenthal. Simplizialzerlegungen von beschränkter Flachheit. *Annals of Mathematics*, 43(3):580–582, 1942. doi:10.2307/1968813.

[GHL25] Moritz Grillo, Christoph Hertrich, and Georg Loho. Depth-bounds for neural networks via the braid arrangement. *Advances in Neural Information Processing Systems*, 38:127800–127818, 2025. doi:10.52202/085713-4257.

[HHL23] Christian Haase, Christoph Hertrich, and Georg Loho. Lower bounds on the depth of integral ReLU neural networks via lattice polytopes. In *International Conference on Learning Representations*, 2023. OpenReview.

[HBD+23] Christoph Hertrich, Amitabh Basu, Marco Di Summa, and Martin Skutella. Towards lower bounds on the depth of ReLU neural networks. *SIAM Journal on Discrete Mathematics*, 37(2):997–1029, 2023. doi:10.1137/22M1489332.

[KMPS23] Christoph Koutschan, Bernhard Moser, Anton Ponomarchuk, and Josef Schicho. Representing piecewise linear functions by functions with small arity. Preprint, arXiv:2305.16933, 2023. arXiv:2305.16933.

[Kuh60] Harold W. Kuhn. Some combinatorial lemmas in topology. *IBM Journal of Research and Development*, 4(5):518–524, 1960. doi:10.1147/rd.45.0518.

[Lu26] Shunqi Lu. Sharp closure thresholds for two-hidden-layer ReLU networks. Preprint, version 1, September 2026. doi:10.5281/zenodo.22728502. Author's PDF.

[RAB+26] Kilian Rueß, Gennadiy Averkov, Florestan Brunck, Moritz Grillo, Christoph Hertrich, Georg Loho, Jack Stade, Moritz Stargalla, Matthew Sun, and Martin Winter. Shallower ReLU network representations via exact linear algebra. Preprint, arXiv:2607.21651, version 2, September 2026. arXiv:2607.21651v2.

[Saf26] Itay Safran. A depth hierarchy for computing the maximum in ReLU networks via extremal graph theory. In *Proceedings of the 39th Conference on Learning Theory*, volume 336 of *Proceedings of Machine Learning Research*, pages 5721–5742, 2026. PMLR 336.

[WB26] Zhimao Wang and Amitabh Basu. Representing MAX functions using two-hidden-layer ReLU networks. Preprint, arXiv:2608.25221, August 2026. arXiv:2608.25221v1.

[WS05] Shuning Wang and Xusheng Sun. Generalization of hinging hyperplanes. *IEEE Transactions on Information Theory*, 51(12):4425–4431, 2005. doi:10.1109/TIT.2005.859246.